

The Compass



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Community Mathematics Centre

Schools of the Krishnamurti Foundation India
(KFI)

Statement of intent

The magazine is published by the Community Mathematics Centre. It is a collaborative initiative of the schools of the Krishnamurti Foundation India (KFI).

Its intention is:

- To share materials for mathematics and science teachers who teach at the high school or middle school levels, and for students who study at these levels.
- To serve as a platform where teachers and students can share their findings and discoveries with others who share their love for the subject.

In addition to mathematical content, in each issue we shall share a few ‘snippets’ from the world of science — short extracts abstracted from articles in online magazines, selected on the basis of intrinsic interest.

The Compass is published as an online magazine free of cost. Articles may be downloaded and used freely by teachers, but with proper attribution.

Edited by Shailesh Shirali, Padmapriya Shirali, Shashidhar Jagadeeshan, Sai Jagan Mohan, Chris John & Swati Sircar on behalf of schools of the KFI

Contact id ... compass.math.2024@gmail.com

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Articles in this issue

I Math & Science Education	1
What's All the Fuss About Patterns?	2
Explaining the Two Patterns	8
Books or Tablets?	13
Child-Led STEAM LAB	18
 II Miscellaneous Articles – 1	 28
Miscellaneous Problems in Number Theory – 2	29
Triangular Numbers – 1	42
Sums of Cubes – 1	56
Fascinating Numbers – 1	67

III Contributed Articles – 1	70
Fast Calculations – 3	71
The Inescapable Friday the 13th	74
Summing the Cubes of the Natural Numbers Using Reversal of Terms	77
Summing the Cubes of the Natural Numbers: Another Approach . .	81
A Conditional Trigonometric Identity	83
Areas of Regions Defined by Circles	86
 IV News From Around the World	 91
Raccoons Solve Puzzles Just for Fun	92
How Horses Whinny	95
Life’s Evil Twins Could Doom Earth	98
Scientists Have a Responsibility to Speak Out	104
The Scientist Who Developed the Polio Vaccine	109
How Lab-Grown Lichen Could Help Us on Mars	112
How AI Can Accelerate Scientific Discovery	117
Polish Researcher Stunned, AI Cracks 20-Year Math Problem	120

Mathematicians Launch First Proof	123
Mathematicians Struggle to Foretell Their Own Future	126
AI Autocomplete Changes How You Think	130
How AI Is Shaping War	133
Economics Has Failed on the Climate Crisis	135
Peeking Inside Neutron Stars	138
What Lies at the Centre of a Galaxy?	142
Ramanujan, Pi and High Energy Physics	146
 V Contributed Articles – 2	 149
The Knight’s Tour Puzzle	150
Symmetric Quadrilaterals: Do We Know Them?	162
The First Transcendental Number	170
A Corollary of the Weak Goldbach Conjecture	175
 VI Culture & History of Mathematics	 180
Katherine Johnson	181

VII Miscellaneous Articles – 2	191
Grandma’s Identities – II	192
A Parabola & Circle Theorem	197
Call for Articles	199

Colour coding

The magazine has articles addressed to teachers who teach at different levels: Lower Primary level; Upper Primary and High School levels; and Senior Secondary and Junior College levels. There is also a Problem Corner. In addition we have the Snippets.

For the convenience of the reader, we use the following colour coding to distinguish between the various levels. Of course, such a division can never be exact. So, our apologies in advance for errors of judgement!

- Primary & Upper Primary (Classes 4, 5, 6, 7)
- Upper Primary & High School (Classes 6, 7, 8, 9, 10)
- Senior Secondary & Junior College (Classes 10, 11, 12)
- Problem Corner
- Snippets

Part I

Math & Science Education

What's All the Fuss About Patterns?

Article contributed by Padmapriya Shirali

Introduction: During my meetings with math teachers, certain questions about recent changes in math curriculum and pedagogy get asked again and again. In these articles, I plan to share my views on these common queries and doubts.

As we go along, we will also explore emerging changes in approaches to teaching, the rationale and a broader vision for math education.

Shifts taking place: Traditionally, primary school mathematics has focused mainly on arithmetic operations and basic mastery, with geometry limited to the study of shapes, properties and terminology; these were seen as sufficient for later study. However, constructive, exploratory and investigative approaches with emphasis on real life applications are gradually being brought into the curriculum and textbooks.

I have observed that many teachers feel uneasy about these shifts. They do not see the need for change and are unsure about how to implement these changes.

The gap: One major ground-level challenge I have observed is this. While high-level goals for math education get referenced in documents like the National Curriculum Framework and the National Education Policy, the vision and its implications rarely reach classrooms. This gap stems from inadequate attention paid to teacher education programmes and lack of initiative on the part of schools to adopt progressive methods and support teacher professional development. Combined with the human resistance to trying out new approaches, it perpetuates outdated practices. This results in lost opportunities in adopting practices that can enhance the understanding and enjoyment of mathematics.

My observation is that the situation is not very different in government schools and private schools. While schools with financial resources are able to acquire math kits for a math lab and digital resources for computer labs, and begin to implement a few activities, the approaches remain piecemeal and lack a comprehensive coherent vision.

“Mathematics is the study of patterns”

An important perspective that has entered key documents in mathematics education — e.g., NCTM Principles and Standards (NCTM, 2000) and NCF-NEP — is that Mathematics is in large part, the “study of patterns”.

This way of looking at mathematics seems to have been made first by the British mathematician, G. H. Hardy, when he wrote, in [3],

A mathematician, like a painter or a poet, is a maker of patterns.

How do mathematics teachers look at such a statement? Do they grasp its significance and its implications for classroom teaching at various levels in school?

Or is it perceived as a new trend about which they (and most others) are unclear with regard to its rationale?

Patterns and rhythms of life: All of us notice the natural patterns of life — daily events like sunrise, sunset, seasonal patterns — in the way our body functions, in activities like cooking, daily travel, etc. Some of these are short-term patterns, while others are long-term patterns.

We also see patterns around us in nature, in ocean waves, in leaves, etc. and in man-made objects like tiles, fabric designs and architecture.

However, we may not see the hidden patterns and their significance in the subjects we teach, in the technological devices that we use, and in the concepts that permeate our daily lives: medicine, transport, production, archaeology, spread of virus, economic trends, planetary movements, etc.

What is involved in the study of patterns? In this context let us look closely at what is involved in the study of patterns. What is the characteristic of patterns? A pattern is a regularity in the world, in human-made design, or in abstract ideas. Patterns follow rules. Rules help us to predict what comes next or what happened in the past. By generalising (abstracting) these rules and modelling we are able to describe how these patterns work. Just as a detective eliminates irrelevant factors and focuses on the crucial data, a mathematician looks at a complex, seemingly chaotic situation to

zone in on the underlying pattern which can then be expressed as a simple, abstract rule. It is a search for a unifying concept and a framework that can handle many instances.

Observation and exploration in science: In the study of science, we encourage students to observe, experiment and gain understanding by going through the investigative process that a scientist follows.

Observation and exploration in mathematics: Do we have a similar vision for mathematics? A mathematician observes, identifies patterns, models, generalises, explains, proves things. Do we provide opportunities for our students to experience such a process? Should we not be doing so?

My experience has been that students enjoy being 'pattern detectives' and feel great delight when they spot a pattern. The delight turns into elation when they are able to reason things out and explain why the pattern 'works'.

All of us who have gone through such a process find it to be a hugely satisfying experience. Within each human being there is an instinctive desire to make sense of things and feel delight in finding the truth of something, however limited it might be. No wonder that the sages who saw the ultimate truth are said to experience bliss!

There is no fun in writing about investigating patterns without actually tasting them! Here are two such problems I shared with a group of teachers recently, during a workshop. We had a lot of fun exploring them. I welcome you to experience it too!

Case Study – 1: This problem is appropriate for primary school.

Ask the students to spot the pattern and to describe it in their own words. Then ask them to continue the pattern for two more rows.

$$1$$

$$1 + 2 + 1$$

$$1 + 2 + 3 + 2 + 1$$

$$1 + 2 + 3 + 4 + 3 + 2 + 1$$

Find the pattern and explain it.

Notes for the teacher: Ask the students to compute the various sums.

Can they predict the next sum? Do they see any relationship between the sum and the number at the centre? Can they represent the growing pattern with dots? (You could provide them with dot paper.) Can they explain “why” this is happening, in their own words?

Case Study – 2: Examine these equalities:

$$1 + 2 = 3,$$

$$4 + 5 + 6 = 7 + 8,$$

$$9 + 10 + 11 + 12 = 13 + 14 + 15, \text{ etc.}$$

Find the pattern and explain it.

Notes for the teacher: For younger students, the problem can be about spotting the pattern. They could try describing the pattern using their own words.

Students of Grades 8 and above can attempt to generalise it and express what they observe in symbolic language. They could also try to explain or prove the pattern.

The answers are given on page 8.

About the author ...

Padmapriya Shirali has been a math educator for the past four decades, teaching at various levels from Class 1 to Class 10. She has a strong interest in math pedagogy and conducting explorations in maths. She is a member of the NCERT textbook committee. She has worked closely with Shri P.K. Srinivasan, an eminent math educationist, and was part of the Rishi Valley Rural Education team which developed the multi-grade learning material known as 'school in a box.' She may be contacted at padmapriya.shirali@gmail.com

Further reading

- [1] Keith Devlin, *Mathematics: The Science of Patterns*. New York: Scientific American Library, 1994
- [2] Michael N. Fried, "Mathematics as the Science of Patterns" from <https://old.maa.org/press/periodicals/convergence/mathematics-as-the-science-of-patterns-introduction>
- [3] Hardy, G. H. *A Mathematician's Apology*. Cambridge: Cambridge University Press, 1992.

Explaining the Two Patterns

Answers to the pattern problems on page 7.

Case study – 1

The problem is to decode the following pattern and to find a satisfactory explanation for it.

$$1$$

$$1 + 2 + 1$$

$$1 + 2 + 3 + 2 + 1$$

$$1 + 2 + 3 + 4 + 3 + 2 + 1$$

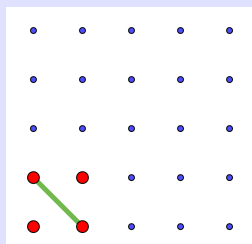
Notes for the teacher: Ask the students to examine the equalities carefully. What do they notice?

What do the sums turn out to be? Square numbers! (If they do not know about square numbers, they may describe it as the “product of a number by itself”, e.g., $1 \times 1 = 1$, $2 \times 2 = 4$, $3 \times 3 = 9$, etc.)

Can they predict the next two sums? (They surely can. The next two sums are 25 and 36.)

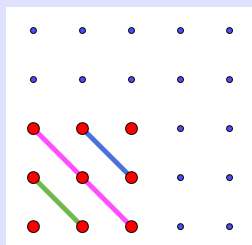
Do they see any relationship between the number at the centre and the sum? (Yes, the sum is the number at the centre multiplied by itself. That is, the sum is the square of the number at the centre.)

Can they represent the growing pattern using dots? Yes, with dot paper:



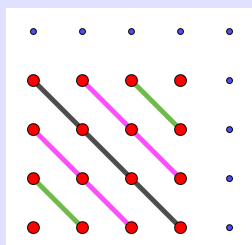
$$1 + 2 + 1 = 2^2$$

Figure 1



$$1 + 2 + 3 + 2 + 1 = 3^2$$

Figure 2



$$1 + 2 + 3 + 4 + 3 + 2 + 1 = 4^2$$

Figure 3

Case study – 2

The problem is to decode the following pattern and to find a satisfactory explanation for it.

$$1 + 2 = 3,$$

$$4 + 5 + 6 = 7 + 8,$$

$$9 + 10 + 11 + 12 = 13 + 14 + 15, \text{ etc.}$$

Notes for the teacher: The participants were asked to examine the equalities carefully. They initially noticed that:

- The number of summands on the LHS was one more than the number of summands on the RHS.
- The sequence of numbers on the RHS continued the sequence of numbers on the LHS.

Does the pattern continue to hold? Participants checked it for the next set of numbers. Is the following true?

$$16 + 17 + 18 + 19 + 20 = 21 + 22 + 23 + 24.$$

It is. And so is this:

$$25 + 26 + 27 + 28 + 29 + 30 = 31 + 32 + 33 + 34 + 35.$$

Could they now generalise the pattern?

Some of them tried to express it in a generalised form using n but found that it was not giving rise to a simple expression as the number sizes and the number of summands on the LHS and RHS were both growing.

Then someone noticed that the starting numbers on the LHS were square numbers, in sequence.

This helped the group come up with a generalised formulation.

Since each row begins with the square of a natural number, the equalities can be expressed as

$$1^2 + (1^2 + 1) = 1^2 + 2,$$

$$2^2 + (2^2 + 1) + (2^2 + 2) = (2^2 + 2 + 1) + (2^2 + 2 + 2),$$

followed by

$$3^2 + (3^2 + 1) + (3^2 + 2) + (3^2 + 3) = (3^2 + 3 + 1) + (3^2 + 3 + 2) + (3^2 + 3 + 3)$$

etc. After reaching this stage, they were able to make quick progress.

In the generalised form, the expression on the LHS is the following:

$$n^2 + (n^2 + 1) + \cdots + (n^2 + n).$$

The expression on the RHS is this:

$$(n^2 + n + 1) + (n^2 + n + 2) + \cdots + (n^2 + n + n).$$

Examine the two expressions. Both sides have the terms $n^2 + 1$, $n^2 + 2$, $\dots n^2 + n$. Subtract these from both LHS and RHS. Here is the result:

$$\text{New LHS} = n^2,$$

$$\text{New RHS} = n \text{ copies of } n \text{ added together} = n \times n = n^2.$$

Both LHS and RHS are equal to n^2 . We have thus arrived at an explanation of the pattern.

Notice the steps involved. We first expressed the pattern in a generalised form. Then we simplified the resulting forms by removing common terms from both sides. Finally, we were left with two expressions clearly equal to each other.

Books or Tablets?

Which medium should schools use for their textual material: physical books or tablets?

Given that life in the future is going to be heavily experienced through the online medium, and given the many obvious advantages that online media offer the user, which physical books cannot offer — the connect with AI, dynamic visualization of various objects, usage of apps such as GeoGebra or browser-based apps such as Desmos, real-time manipulation of data — it seems natural for schools to choose tablets over physical books.

But there has been increasingly a pushback to not move so enthusiastically towards digital devices. This has come from multiple directions, including psychologists, educators and parents. There are many different reasons for this, not least being mental and physical health concerns. There is also the obvious issue of cost.

We invite readers to reflect on the issue.

Meanwhile, here is a summary of an article about what is happening in Sweden with regard to this very issue.

Adapted from: “Why Swedish Schools Are Bringing Back Books.”

From the website <https://undark.org/>. URL of article: <https://undark.org/2026/04/01/sweden-schools-books/>.

Author: Joshua Cohen

Sweden is systematically reintroducing physical books, handwriting, and cell phone-free environments into its educational system after decades of heavy reliance on digital learning tools. This policy shift stems from concerns over declining cognitive skills and represents a global inflection point in educational philosophy regarding the role of technology in classrooms.

The Swedish Policy Pivot

In a dramatic reversal of previous digital-first policies, the Swedish government initiated a “back to basics” approach in 2023 that prioritizes foundational skills like reading and writing in early grades. Educational authorities are replacing tablets with physical media, ensuring that students are “learning to write the old-fashioned way: by hand, with a pencil or pen, on sheets of paper”. This transition includes implementing cell phone-free policies across the country to minimize digital distractions and foster deeper classroom engagement.

Financial Investments in Print

To facilitate this transition, the Swedish education ministry has heavily subsidized the return of analogue materials, allocating \$83 million strictly

for textbooks and teachers' guides in a single year. An additional \$54 million was dedicated to purchasing both fiction and non-fiction literature to build robust physical libraries. The ultimate governmental aim is "for every student to have a physical textbook for each subject" among the nation's 11 million citizens.

Pedagogical Rationale

The decision to abandon screen-heavy instruction was driven by a noticeable decline in Swedish standardized test scores across reading, mathematics, and science between 2000 and 2022. Linnaeus University researcher Linda Fälth explicitly noted that a "broader cultural reassessment" occurred because "concerns emerged about screen time, distraction, reduced deep reading, and the erosion of foundational skills such as sustained attention and handwriting". Proponents of this reform maintain that basic numeracy and literacy must be firmly established first, an objective for which physical books are inherently better suited.

Impact on Cognitive Development

A growing body of research corroborates the Swedish approach, demonstrating that reading on digital screens is significantly more mentally demanding and leads to reduced comprehension, diminished memory retention, and increased eye strain.

Literacy consultant Pam Kastner emphasizes that the brain's cognitive architecture is built for print, arguing succinctly that "Technology is a tool, not a teacher".

Neuroscientist Jared Cooney Horvath has similarly testified that continuous digital immersion has resulted in Generation Z being “less cognitively capable than their parents”.

The American Educational Context

Unlike Sweden, the United States continues to heavily prioritize digital learning, spending \$30 billion in 2024 on educational technology—an expenditure ten times greater than what was spent on traditional textbooks. Professor Naomi Baron attributes this partly to financial incentives, noting that “commercial textbook publishers have been pushing digital materials — heavily for financial reasons generally ignoring the research” regarding reading comprehension. Furthermore, technology giants are actively lobbying American schools to integrate artificial intelligence literacy, ensuring students are prepared for digitally demanding employers.

Resistance to Digitalization

Despite massive institutional investments in technology, grass-roots resistance is forming among American educators and parents who are increasingly concerned about excessive screen time and its potentially addictive distractions. Social psychologist Jonathan Haidt issued a stark warning regarding this trend, stating that “Putting computers and tablets on students desks in K-12 may turn out to be among the costliest mistakes in the history of education”. Consequently, parent networks are organizing to opt out of school-issued devices in favour of evidence-based analogue learning methods that yield superior information retention.

Recalibration Over Reversal

Swedish officials clarify that their mandate is not an absolute rejection of technology, but rather an intentional restructuring of when and how it is deployed. The Swedish Ministry of Education and Research stated that while digitalization remains beneficial, its implementation “must be carried out carefully and thoughtfully”. By choosing to “judiciously establish boundaries,” educators intend to introduce digital tools sequentially and only after essential foundational skills are successfully mastered.

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Gemini 3.1 Pro/Perplexity* LLM.

A Child-Led STEAM LAB

Article contributed by Pramod Maithil

There is a deceptively simple idea at the heart of a well-designed STEAM Lab: put a child's joy first, and learning follows. Not as a coincidence, but as a natural consequence. When children are deeply engaged and genuinely happy in what they are doing, they explore more deeply, persist through difficulty, and develop skills almost without realising it.



This is the central conviction of the Natural Learning Model — a pedagogical framework that places the child's happiness at the centre, with diverse domains of exploration radiating outward.

Figure 4

In this model, learning outcomes are not targets to be pursued. They are by-products of genuine engagement.

The central point represents the happiness of the child. The squares radiating outward represent different domains of exploration. And the circle that emerges around them in the middle — the aspired learning outcomes — arises naturally from the whole arrangement.

“Most learning is not the result of instruction. It is rather the result of unhampered participation in a meaningful setting.” — Ivan Illich

A STEAM Lab built on this conviction looks and feels very different from a conventional learning space. It is not a room where children execute pre-designed experiments or follow step-by-step instructions. It is a space where children choose what to make, decide how to make it, manage the resources they need, and present their work to their peers — with adults available on demand, but never in the driver’s seat.

Head, Heart & Hand — A Holistic Approach

Mahatma Gandhi’s vision of holistic education, rooted in the development of the Head, Heart, and Hand, provides the philosophical spine of the STEAM Lab. These are not three separate goals — they are three dimensions of the same experience, activated simultaneously when a child is genuinely absorbed in their work.

Head: The Head represents ideas, imagination, and critical thinking. When a child works on a project, they consider possibilities, anticipate challenges, and use reasoning to guide their creative decisions. The cognitive work is real — but it emerges from curiosity, not from instruction.

Heart: The Heart begins with joy — the deep satisfaction that comes from making something meaningful. Along the way, children develop emotional resilience and self-motivation. They experience the excitement of a new idea, the uncertainty of trial and error, and the pride of seeing their vision take form. Even when things don't work, they learn to adapt, persist, and find value in the journey itself.

Hand: The Hand is about translating thought into action — developing practical skills with real tools and materials. Whether cutting wood, wiring a circuit, mixing dyes for batik, or sculpting clay, the act of doing reinforces understanding and builds the confidence to turn an idea into something tangible.

When Head, Heart, and Hand work in unison, children are not just learning subject matter — they are developing as whole human beings. Observations in STEAM Labs running on this model consistently show near-100% engagement: children do not clock-watch when they are genuinely absorbed.

Child-Led STEAM: The Democratic Engine

What gives the STEAM Lab its structural backbone is not the equipment or the spaces — it is the fact that the Lab is run, in large part, by children themselves. This is not symbolic participation. It is genuine, democratic, operational ownership.

At the start of each year, the Lab holds a local election. Students campaign, make their case to their peers, and are voted into roles across three teams that manage the Lab's core functions. The teacher is freed to focus entirely

on academic facilitation — asking the right question, offering technical guidance, inspiring students by working on their own projects alongside them.



Figure 5

The Three Teams

The **Idea Team** manages the Planning Board. Every student who wants to start a project must first present their idea to the Idea Team, who ask questions, offer feedback, and register the project on the board for the whole community to see.

Once written up, a project becomes a commitment — a promise made in public. This simple act eliminates abandoned ideas and creates a culture of follow-through.

The **Bank Team** runs the Lab's currency system — overseeing transactions, recording balances, and issuing "loans" for ambitious projects.

The **Material Team** organises, stores, and distributes tools and resources, ensures fair access, and maintains the Lab's physical order.

The Currency System: An Entrepreneur Pathashaala

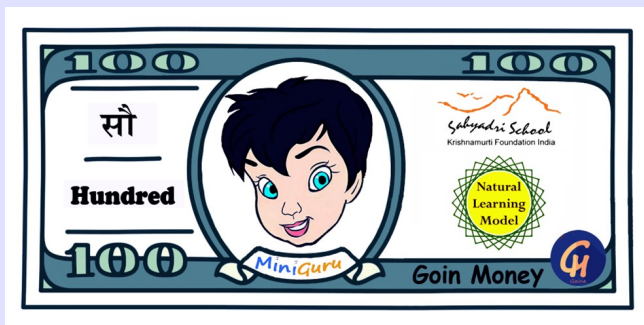


Figure 6: Currency System of the STEAM Lab

One of the distinctive and powerful features of the STEAM Lab is its currency system — a specially created dummy currency that children use to buy materials and rent tools. When a project is complete, it is brought to the Podium, where peers evaluate it for originality, usefulness, and craftsmanship, and it is sold back to the Lab at an agreed price.

The effect is profound. A child who would previously take a full bottle of glue for a small task now pauses to estimate exactly how much they need. Resource awareness, financial planning, and the habit of valuing one's own work are learned not through lessons, but through practice.



Alongside the currency system, the Planning Board acts as the Lab’s shared roadmap.

Before starting, every student writes their project on the board after presenting to the Idea Team.

The board makes the whole community’s work visible — and once a project is on it, it becomes a social commitment that motivates follow-through.

Figure 7: The STEAM Lab Planning Board

Together, these three systems: The democratic election, the currency system, and the Planning Board — form the operational backbone of a Child-Led STEAM Lab. They are not add-ons. They are the architecture of the whole experience. Without them, the Lab is just a well-stocked room.

The Spaces That Make It Possible

The physical environment is designed to draw children into exploration. Materials are stored openly — in racks, labelled boxes, and open baskets — so that children can see what is available and let materials spark new ideas. Reusable materials — cardboard tubes, offcuts of wood and cloth, broken

appliances, salvaged parts — supplement what is purchased. A school's waste-sorting system, thoughtfully organised, becomes one of the Lab's richest resources.

A fully realised STEAM Lab is not a single room but a network of creative spaces, each with its own character and possibilities, interconnected so that a child's project can move fluidly between them. The spaces are in constant conversation: a batik panel created in the studio finds its way onto a structure built in carpentry; a circuit wired in the STEAM Room brings light to a clay sculpture.

The STEAM Room is the central hub for electrical and electronic work, where the Planning Board and Lab Bank live. Computers sit alongside wires, LEDs, motors, switches, and sensors.

- The [Tinkershade](#) is a mechanical playground of nuts, bolts, gears, pulleys, and wheels — where ideas take on movement and structure.
- The [Carpentry Workshop](#) offers the deep satisfaction of working with wood: measuring precisely, sawing cleanly, joining firmly, finishing with care.
- The [Batik Studio](#) brings the ancient art of wax-resist dyeing into conversation with modern design, often integrating fabric panels into carpentry structures.
- The [Art Room](#) bursts with colour, texture, and mixed media.
- The [Clay Modelling Area](#) offers the grounding experience of working with earth itself — producing functional pots, sculptures, or textured components in larger pieces.
- [Science Laboratories — Physics, Chemistry, Biology](#) — also open their doors to STEAM projects, so a carpentry structure can be tested

for load-bearing strength, or a dye recipe explored as a chemical experiment.

Not every school can build all these stations at once, so having relevant resources inside the STEAM room is sufficient. A Lab with two or three strong stations, a functioning currency system, an active Planning Board, and student-run teams will outperform a fully-equipped room that is teacher-directed. The strength of this model lies in its systems and processes, not in the size of the space.

Lab and Classroom: Two Sides of One Education



The STEAM Lab is not a replacement for classroom learning — rather, it is its complement. In the classroom, learning follows a planned sequence; in the Lab, students apply, extend, and connect what they know in open-ended ways.

Figure 8: Difference between a STEAM Lab and a classroom

The teacher's role thus shifts from director to companion — stepping in when needed to pose a thought-provoking question, suggest a material or technique, or inspire by working on their own project alongside the students.

A distinctive feature is how students learn from one another. The energy in the room comes from informal exchanges — watching a peer test a design, discussing a challenge, combining ideas from different projects.



This natural peer-to-peer learning is both powerful and organic.

The Lab is designed to encourage this kind of learning.

Figure 9: Peer-to-peer learning in the Lab

Learning Through Failure. Growing Through Making.

In many settings, projects are judged only by whether they “succeed.” In the STEAM Lab, the measure is broader. A working prototype is certainly a win, but so is a failed attempt that taught new techniques, or the decision to abandon a first plan in favour of a better idea. By removing the fear of failure, the Lab opens space for risk-taking, experimentation, and honest reflection.

This mindset builds resilience — the ability to face setbacks without losing confidence. It encourages a healthier, more adventurous approach to learning. And because children manage their own resources, plan their own projects, and govern their own space, they develop something that no

curriculum can directly teach: a sense that they are capable, that their ideas have value, and that they can build something meaningful in the world.

The Lab is not just about projects or products. It is about nurturing a way of being in the world — one that blends critical thought with creativity, joy with responsibility, and imagination with the courage to act.

About the author

PRAMOD MAITHIL is an educator with 25+ years of hands-on experience working with children, teachers, and schools. He spent 12 years at Eklavya focusing on learning and curriculum. He promotes natural learning practices and has designed and set up multiple T-LAB (STEAM) spaces for creativity in schools across India. He is currently a Science and STEAM Educator at Sahyadri School.

He may be contacted at pramod.maithil@sahyadrischool.org.

Part II

Miscellaneous Articles – 1

Miscellaneous Problems in Number Theory – 2

As in the previous issue, we study here a few miscellaneous problems in number theory. We state the problems first and present the solutions later. We urge you to try solving the problems on your own before studying the solutions.

Problems

Problem 1. Find all positive integers n such that $n(n + 2026)$ is a perfect square.

Problem 2. Find all triples (a, b, c) of integers such that

$$a^2 + b^2 + c^2 - bc - ca - ab = 13.$$

Problem 3. Find the last four digits of the product of all odd numbers less

than or equal to 2025, i.e., of the number

$$1 \cdot 3 \cdot 5 \cdot 7 \cdot \dots \cdot 2021 \cdot 2023 \cdot 2025.$$

Problem 4. Find all positive integers n for which the number

$$\frac{1}{n(2n+1)}$$

has a terminating decimal expansion.

Problem 5. Prove that every Fibonacci number that is divisible by 11 is also divisible by 5.

Problem 6. Prove that every Fibonacci number that is divisible by 31 is also divisible by 61.

Problem 7. Find all pairs of natural numbers m, n such that $1 + 5 \cdot 2^m = n^2$.

Problem 8. Prove the following rule — discovered in the 9th century by the Arab mathematician Thabit ben Korrah — for generating pairs of **amicable numbers**. Let n be an integer greater than 1, such that the numbers

$$p = 3 \cdot 2^{n-1} - 1, \quad q = 3 \cdot 2^n - 1, \quad r = 9 \cdot 2^{2n-1} - 1$$

are all prime numbers. Then $\{2^n pq, 2^n r\}$ are a pair of amicable numbers.

Solutions

Solution to Problem 1

To find all positive integers n such that $n(n + 2026)$ is a perfect square.

Let $n(n + 2026) = m^2$, where m is a positive integer. Then we have,

$$n^2 + 2026n = m^2,$$

$$\therefore n^2 + 2026n + 1013^2 = m^2 + 1013^2, \quad (\text{by “completing the square”})$$

$$\therefore 1013^2 = (n + 1013)^2 - m^2,$$

$$\therefore 1013^2 = (n + 1013 + m) \times (n + 1013 - m).$$

Since 1013 is a prime number, the only factorization of 1013^2 into two unequal numbers is $1013^2 = 1013^2 \times 1$. Hence we must have,

$$n + 1013 + m = 1013^2,$$

$$n + 1013 - m = 1.$$

The equations yield $n = 512072$ and $m = 513084$, so the only solution in positive integers to the given equation is $n = 512072$, with

$$512072 \times (512072 + 2026) = 513084^2.$$

Remark

From the above, we deduce the following corollary. If p is an odd prime, then there is precisely one positive integer n such that $n^2 + 2pn$ is a perfect square, namely $n = \frac{1}{2}(p - 1)^2$.

Solution to Problem 2

We need to find all integer triples (a, b, c) such that

$$a^2 + b^2 + c^2 - bc - ca - ab = 13.$$

There is no loss of generality in assuming that $a \geq b \geq c$. We can relax this assumption later.

You may have seen this problem earlier: **Show that if a, b, c are real numbers such that $a^2 + b^2 + c^2 - bc - ca - ab$, then $a = b = c$.** The standard way to solve this problem is to write the expression $a^2 + b^2 + c^2 - bc - ca - ab$ as a sum of squares. It is easy to verify that

$$(a - b)^2 + (b - c)^2 + (c - a)^2 = 2(a^2 + b^2 + c^2) - 2(bc + ca + ab).$$

We make use of the same identity for our problem. We have,

$$a^2 + b^2 + c^2 - bc - ca - ab = 13,$$

$$\therefore (a - b)^2 + (b - c)^2 + (c - a)^2 = 26.$$

We see that 26 has been written as a sum of squares of non-negative integers. What are all the ways of doing this? It is easy to verify that the only way is the following:

$$26 = 16 + 9 + 1 = 4^2 + 3^2 + 1^2.$$

Making use of $a \geq b \geq c$, we infer that $(a - b, b - c, a - c)$ is a permutation of $(4, 3, 1)$. Since $a - c = (a - b) + (b - c)$, the only possibilities are

$$(a - b, b - c, a - c) = (1, 3, 4)$$

and

$$(a - b, b - c, a - c) = (3, 1, 4).$$

Both possibilities yield infinitely many solutions. Treating c as a parameter, the first possibility yields $(a, b, c) = (c + 4, c + 3, c)$ while the second possibility yields $(a, b, c) = (c + 4, c + 1, c)$. Here, c can take any integer value.

Conclusion: The integer triples (a, b, c) with $a \geq b \geq c$ and $a^2 + b^2 + c^2 - bc - ca - ab = 13$ are all of the form $(k + 4, k + 3, k)$ or $(k + 4, k + 1, k)$ for some integer k .

The required answer thus consists of all permutations of these triples.

Solution to Problem 3

We are required to find the last four digits of the product

$$1 \cdot 3 \cdot 5 \cdot 7 \cdot \dots \cdot 2021 \cdot 2023 \cdot 2025.$$

Let N denote the given number. We must find the remainder in the division $N \div 10000$. Since $10000 = 5^4 \times 2^4$, and N is visibly divisible by 5^4 , it suffices to find the remainder in the division $N \div 16$.

It is easy to check, without actual multiplication, that

$$1 \cdot 3 \cdot 5 \cdot 7 \cdot 9 \cdot 11 \cdot 13 \cdot 15 \equiv 1 \pmod{16}.$$

For we have, modulo 16,

$$1 \cdot 15 \equiv -1, \quad 3 \cdot 5 \equiv 11 \times 13 \equiv -1, \quad 7 \cdot 9 \equiv -1,$$

and so

$$1 \cdot 3 \cdot 5 \cdot 7 \cdot 9 \cdot 11 \cdot 13 \cdot 15 \equiv (-1)^4 \equiv 1 \pmod{16}.$$

We deduce that

$$17 \cdot 19 \cdot 21 \cdot 23 \cdot 25 \cdot 27 \cdot 29 \cdot 31 \equiv 1 \pmod{16},$$

$$33 \cdot 35 \cdot 37 \cdot 39 \cdot 41 \cdot 43 \cdot 45 \cdot 47 \equiv 1 \pmod{16},$$

and so on, and this goes on till

$$2001 \cdot 2003 \cdot 2005 \cdot 2007 \cdot 2009 \cdot 2011 \cdot 2013 \cdot 2015 \equiv 1 \pmod{16}.$$

Hence

$$1 \cdot 3 \cdot 5 \cdots \cdots 2013 \cdot 2015 \equiv 1 \pmod{16}.$$

So we only need to consider the factors beyond 2015. Now, modulo 16,

$$2017 \equiv 1, \quad 2019 \equiv 3, \quad 2021 \equiv 5, \quad 2023 \equiv 7, \quad 2025 \equiv 9,$$

and

$$1 \cdot 3 \cdot 5 \cdot 7 \cdot 9 \equiv 1 \cdot (-1) \cdot (-1) \equiv 1 \pmod{16}.$$

It follows that

$$N \equiv 1 \pmod{16}.$$

Hence

$$N \equiv 1 \pmod{16}, \quad N \equiv 0 \pmod{625}.$$

Since $625 \equiv 1 \pmod{16}$, it follows that the last four digits of N are 0625.

Solution to Problem 4

We have been asked to find all positive integers n for which the number

$$\frac{1}{n(2n+1)}$$

has a terminating decimal expansion. It is known that the fraction $1/k$ has a terminating decimal expansion if and only if k has no prime factors other than 2 or 5. So an equivalent formulation of the problem is this: Find all positive integers n for which the number $n(2n + 1)$ has no prime factors other than 2 or 5. (Obviously, the same property must be true of both n and $2n + 1$ individually.)

Here is one number with this property: $n = 2$, with $n(2n + 1) = 10$. We must ascertain whether there are any more such numbers.

Let n have the stated property. We deduce the following.

- $2n + 1$ is odd, so it must have no prime factor other than 5. Hence $2n + 1 = 5^a$ for some integer $a \geq 1$.
- Since n and $2n + 1$ are coprime, it follows from the above that n is not divisible by 5. Hence n is a power of 2. The same is true of $2n$. Hence $2n = 2^b$ for some positive integer b .
- So we have $2n = 2^b$ and $2n + 1 = 5^a$, and therefore, $5^a - 2^b = 1$.
- So we must look for all instances where a power of 5 exceeds a power of 2 by 1. We already have one such instance at hand: $5^1 = 5$ and $2^2 = 4$. We must find out whether there are any others.

Analysis: Let a and b be positive integers such that $5^a - 2^b = 1$. Then:

- b must be even. (If b were odd, then $2^b + 1$ would be divisible by 3, hence 5^a would be divisible by 3, but 5^a cannot be divisible by 3.)
- a must be odd. (If a were even, then $5^a - 1$ would be divisible by 6, hence 2^b would be divisible by 3; but 2^b cannot be divisible by 3.)

- Suppose that $b \geq 3$. Then, reading the equality modulo 8, we would have $5^a \equiv 1 \pmod{8}$.
- Now, modulo 8, the powers of 5 go through the following cycle: 5, 1, 5, 1, ... That is, $5^1 \equiv 5$, $5^2 \equiv 1$, $5^3 \equiv 5$, ...; note that the 1's come in every second place.
- It follows from the above that a must be even.
- We have reached a pair of contradictory conclusions: that a must be odd, and that a must be even. The contradiction clearly stems from the assumption that there is a solution with $b \geq 3$. It follows that there can be no such solution.
- Hence the only solution is $a = 1, b = 2$. This means that the only positive integer n with the property that the fraction

$$\frac{1}{n(2n+1)}$$

has a terminating decimal expansion is $n = 2$.

Solution to Problem 5

We must show that every Fibonacci number that is divisible by 11 is also divisible by 5. Recall the definition of the Fibonacci numbers, $\{F_n\}_{n \geq 0}$:

$$F_0 = 0, \quad F_1 = 1, \quad F_n = F_{n-1} + F_{n-2} \text{ for all } n > 1.$$

Applying the definition recursively, we obtain the following numbers:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
F_n	1	1	2	3	5	8	13	21	34	55	89	144	233	377	610

To prove the stated property, we use the following fundamental property of the Fibonacci sequence: $\text{If } m \mid n, \text{ then } F_m \mid F_n$.

Sequences with such a property are called *divisibility sequences*.

The following instances illustrate what this means: $F_4 \mid F_8$; $F_6 \mid F_{12}$; $F_7 \mid F_{14}$; and so on.

If we read the Fibonacci sequence modulo 11, here is what we obtain:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14
$F_n \pmod{11}$	1	1	2	3	5	8	2	10	1	0	1	1	2	3

We see that the Fibonacci numbers mod 11 go through a cycle of length 10:

1, 1, 2, 3, 5, 8, 2, 10, 1, 0 (the cycle repeats after this)

Examining the above cycle, we deduce the following:

$$11 \mid F_n \text{ precisely when } 10 \mid n.$$

Similarly, if we read the Fibonacci sequence modulo 5, we obtain:

1, 1, 2, 3, 0, 3, 3, 1, 4, 0, 4, 4, 3, 2, 0, ...,

and we arrive at the following:

$$5 \mid F_n \text{ precisely when } 5 \mid n.$$

Hence we have,

$$11 \mid F_n \implies 10 \mid n \implies 5 \mid n \implies 5 \mid F_n,$$

so if a Fibonacci number is divisible by 11, then it is also divisible by 5.

Solution to Problem 6

We must show that every Fibonacci number that is divisible by 31 is also divisible by 61. We leave this as an exercise for you. It may be solved in the same manner as earlier (but the numbers involved are much larger).

Solution to Problem 7

We are required to find all pairs of natural numbers m, n such that

$$1 + 5 \cdot 2^m = n^2.$$

The equation may be rewritten as

$$5 \cdot 2^m = (n - 1)(n + 1).$$

The equation implies that one quantity in $\{n - 1, n + 1\}$ is of the form $5 \cdot 2^a$ and the other quantity is of the form 2^b . We consider both possibilities.

First possibility: Suppose that $n - 1 = 5 \cdot 2^a$ and $n + 1 = 2^b$. Then

$$2^b - 5 \cdot 2^a = 2,$$

hence $a < b$ and

$$2^a (2^{b-a} - 5) = 2.$$

This leads to

$$2^{b-a} - 5 = 1, \quad \therefore 2^{b-a} = 6,$$

which clearly is not possible. Therefore, this possibility does not work out.

Second possibility: We therefore have that $n + 1 = 5 \cdot 2^a$ and $n - 1 = 2^b$. Hence

$$5 \cdot 2^a - 2^b = 2.$$

Suppose that $a > b$. Then we would have

$$2^b (5 \cdot 2^{a-b} - 1),$$

leading to

$$5 \cdot 2^{a-b} - 1 = 1, \quad \therefore 5 \cdot 2^{a-b} = 2,$$

which again is not possible.

Therefore we must have $a \leq b$. This now leads to

$$2^a (5 - 2^{b-a}) = 2.$$

In turn, this leads to

$$2^a = 2, \quad \therefore a = 1,$$

and

$$5 - 2^{b-a} = 1, \quad \therefore 2^{b-a} = 4, \quad \therefore b - a = 2, \quad \therefore b = 3.$$

It follows that $m = a + b = 4$ and $n = 9$. It is easy to back-check and verify that this fits the given equation.

The given equation thus has just one solution, $(m, n) = (4, 9)$.

Solution to Problem 8

We must show that if $n > 1$ is such that the numbers

$$p = 3 \cdot 2^{n-1} - 1, \quad q = 3 \cdot 2^n - 1, \quad r = 9 \cdot 2^{2n-1} - 1$$

are all primes, then $\{2^n pq, 2^n r\}$ are a pair of amicable numbers. This means that we must show the following:

Sum of the proper factors of $2^n pq$ is $2^n r$,

Sum of the proper factors of $2^n r$ is $2^n pq$.

Let us compute the two sums involved. We use a basic result about the sum-of-divisors function (reference given below).

Sum of the proper factors of $2^n pq$

$$\begin{aligned}
 &= (2^{n+1} - 1) \cdot (1 + p) \cdot (1 + q) - 2^n pq \\
 &= (2^{n+1} - 1) \cdot 3 \cdot 2^{n-1} \cdot 3 \cdot 2^n - 2^n (3 \cdot 2^{n-1} - 1) \cdot (3 \cdot 2^n - 1) \\
 &= 9 \cdot 2^{3n-1} - 2^n \\
 &= 2^n \cdot r.
 \end{aligned}$$

In the same way,

Sum of the proper factors of $2^n r$

$$\begin{aligned}
 &= (2^{n+1} - 1) \cdot (1 + r) - 2^n r \\
 &= (2^{n+1} - 1) \cdot 9 \cdot 2^{2n-1} - 2^n (9 \cdot 2^{2n-1} - 1) \\
 &= 9 \cdot 2^{3n-1} - 9 \cdot 2^{2n-1} + 2^n \\
 &= 2^n \cdot pq,
 \end{aligned}$$

as required.

Sum-of-divisors theorem

The main results used in the above derivation are listed below.

Result:–1 If m, n are coprime positive integers, then

Sum of all the divisors of mn

= Sum of all the divisors of $m \times$ Sum of all the divisors of n .

This result implies the next one.

Result:–2 Let the prime factorization of n be

$$n = p^a \times q^b \times r^c \times \dots,$$

where $p, q, r, \dots$ are distinct primes. Then the sum of all the divisors of n is equal to

$$\begin{aligned} & \left(1 + p + p^2 + \dots + p^a\right) \\ & \times \left(1 + q + q^2 + \dots + q^b\right) \\ & \times \left(1 + r + r^2 + \dots + r^c\right) \times \dots \end{aligned}$$

Example: Take $n = 200 = 2^3 \times 5^2$. According to the above, the sum of the divisors of 200 is

$$(1 + 2 + 2^2 + 2^3) \times (1 + 5 + 5^2) = 15 \times 31 = 465.$$

The reader should verify that this is correct.

Tales about Triangular Numbers – 1

Triangular numbers are amazingly easy to define and even visualize, and they have many beautiful properties. We describe some of their features in this series of articles.

Introducing the triangular numbers

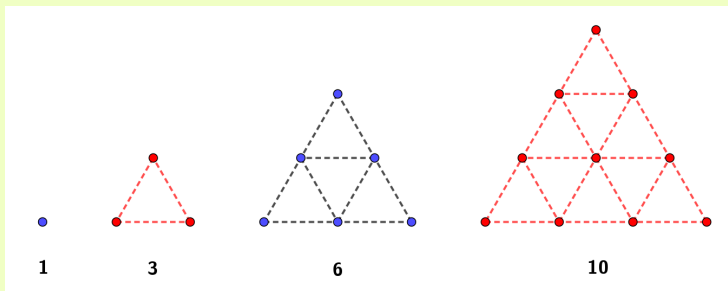


Figure 10

The dot figures sketched above indicate what the term **triangular number** means. Continuing, we obtain an unending number sequence,

$$1, 3, 6, 10, 15, 21, 28, 36, 45, 55, \dots$$

We define T_n to be the n -th number in this sequence (' T ' for "triangular"):

$$T_1 = 1, \quad T_2 = 3, \quad T_3 = 6, \quad T_4 = 10, \quad \dots$$

We can define the T -numbers in a purely arithmetical manner as well:

$$T_1 = 1,$$

$$T_2 = 1 + 2 = 3,$$

$$T_3 = 1 + 2 + 3 = 6,$$

$$T_n = 1 + 2 + 3 + \dots + n = \sum_{k=1}^n k.$$

Here is a recursive definition:

$$T_1 = 1, \quad T_n = T_{n-1} + n \quad \text{for } n > 1.$$

Properties of the triangular numbers

Considering how simple their definition is, the sheer variety of properties and identities that the triangular numbers obey is astonishing. We shall study as many of them as seems feasible. The reader will need to continually refer to the table of values given below. (The table may be prepared using the above recursive definition repeatedly.)

T\U	0	1	2	3	4	5	6	7	8	9
0	0	1	3	6	10	15	21	28	36	45
1	55	66	78	91	105	120	136	153	171	190
2	210	231	253	276	300	325	351	378	406	435
3	465	496	528	561	595	630	666	703	741	780
4	820	861	903	946	990	1035	1081	1128	1176	1225
5	1275	1326	1378	1431	1485	1540	1596	1653	1711	1770
6	1830	1891	1953	2016	2080	2145	2211	2278	2346	2415
7	2485	2556	2628	2701	2775	2850	2926	3003	3081	3160
8	3240	3321	3403	3486	3570	3655	3741	3828	3916	4005
9	4095	4186	4278	4371	4465	4560	4656	4753	4851	4950

Using the table we get $T_{27} = 378$, $T_{43} = 946$, $T_{74} = 2775$, etc.

A formula for the n-th triangular number

Let's start with the most basic formula: the generating formula of the sequence. That is, a formula for T_n explicitly in terms of n .

A simple way to guess the formula is to list the values of $T_n \div n$:

n	1	2	3	4	5	6	7	8	9	10
T_n	1	3	6	10	15	21	28	36	45	55
$T_n \div n$	1	$1\frac{1}{2}$	2	$2\frac{1}{2}$	3	$3\frac{1}{2}$	4	$4\frac{1}{2}$	5	$5\frac{1}{2}$

Noticing the repeated occurrence of $\frac{1}{2}$ in the bottom row, we are prompted to double all the entries. So let us list the values of $2T_n \div n$ instead. Here is what we get.

n	1	2	3	4	5	6	7	8	9	10
T_n	1	3	6	10	15	21	28	36	45	55
$2T_n \div n$	2	3	4	5	6	7	8	9	10	11

The formula now jumps out at us: $2T_n \div n = n + 1$, or

$$T_n = \frac{n(n+1)}{2}.$$

This seems to fit the table perfectly. Let us now prove the formula. There are several ways of doing so.

Reversing the summands: Gauss's approach

This is the most obvious approach and it is linked to a well-known story about the mathematician Carl F Gauss when he was a young boy. (Note: It is possible that the story is a myth and is simply part of mathematical folklore!)

All we do is take *two copies* of the sum that defines T_n , reverse the order of

the summands in one copy, and then pair up the terms:

$$T_n = 1 + 2 + 3 + \cdots + n-1 + n$$

$$T_n = n + n-1 + n-2 + \cdots + 2 + 1$$

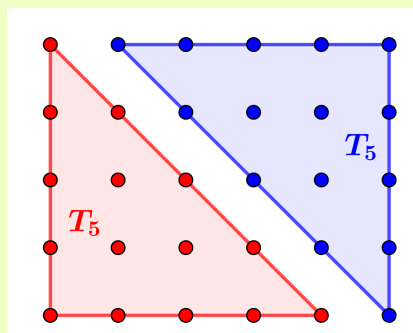
Pairing up the terms “vertically” and adding, we get

$$2T_n = \underbrace{(n+1) + (n+1) + \cdots + (n+1)}_{n \text{ copies}} = n \times (n+1),$$

$$\therefore T_n = \frac{n(n+1)}{2}.$$

A visual proof

The idea of pairing up the first number with the last number, the second with the second last, and so on, may also be expressed in a visual form. The figure illustrates this for the case $n = 5$.



The red triangle and the blue triangle are congruent copies of each other and are representations of T_5 . Observe how we have paired $1 + 5$, $2 + 4$, etc. The two grids together form a 5×6 grid, giving $2T_5 = 5 \times 6$. The extension to the general case should be clear.

Figure 11: Visually illustrating the proof by reversal of the terms

Proof using telescoping

Another approach used to prove the identity is *proof by telescoping*. This is based on the following claim. Define the function f on the integers by

$$f(n) := \frac{n(n+1)}{2}.$$

Then we have:

$$f(n) - f(n-1) = n.$$

The claim is trivial to verify using straightforward algebra.

Now all we have to do is write the corresponding relations for $n, n-1, n-2, \dots, 2, 1$ and then add the corresponding sides of all the relations:

$$f(n) - f(n-1) = n,$$

$$f(n-1) - f(n-2) = n-1,$$

$$f(n-2) - f(n-3) = n-2,$$

$$\dots \dots = \dots,$$

$$f(2) - f(1) = 2,$$

$$f(1) - f(0) = 1.$$

When we add the corresponding sides, a mass cancellation occurs (please check!), and we are left with:

$$f(n) - f(0) = n + (n-1) + (n-2) + \dots + 2 + 1.$$

Since $f(0) = 0$, this amounts to saying that

$$1 + 2 + \dots + n = f(n),$$

i.e.,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof by telescoping is a powerful and frequently used idea in algebra.

Remark: Yet another approach, which we do not describe here, is *proof by induction*. This is essentially based on the same identity as the proof by telescoping. However, they are generally presented as two completely different proofs.

Connection with the Pascal Triangle

You are no doubt familiar with the Pascal Triangle. (Who isn't?) Here it is:

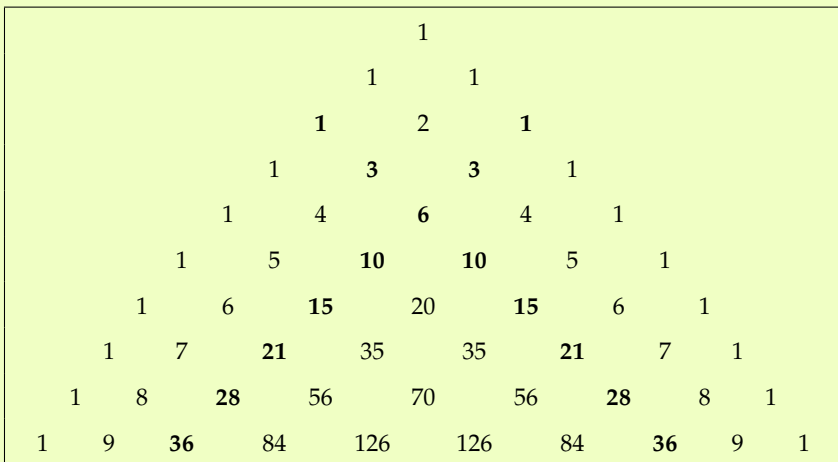


Figure 12: Pascal's triangle. Do you see the triangular numbers?

The presence of the triangular numbers in the Pascal triangle is not a coincidence. Rather, it follows directly from the defining property of the Pascal triangle. Do you see why?

The above figure highlights an extremely important connection between the triangular numbers and the combinatorial coefficients. We shall more about this connection in the second part of this article.

Familiar identities

Here are some of well known identities that the triangular numbers obey.

Connection with the square numbers (there are many of them!):

1. $T_{n-1} + T_n = n^2$ Example: $T_3 + T_4 = 6 + 10 = 16 = 4^2$

The algebraic verification of this is trivial; we do not need to show the details. But it is fun to prove this identity visually.

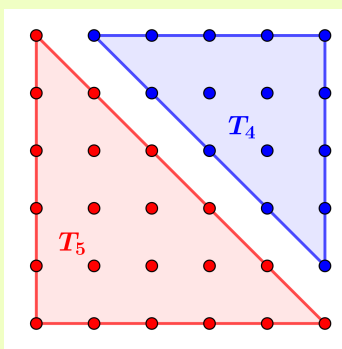


Figure 13: Visual demonstration of the equality $T_4 + T_5 = 5^2$

2. $8T_n + 1 = (2n + 1)^2$ Example: $8T_3 + 1 = (8 \times 6) + 1 = 49 = 7^2$

For a visual proof, we first take two copies of T_n and join them to make a $n \times (n + 1)$ rectangle. Then we take four copies of this rectangle and join them together as shown below.

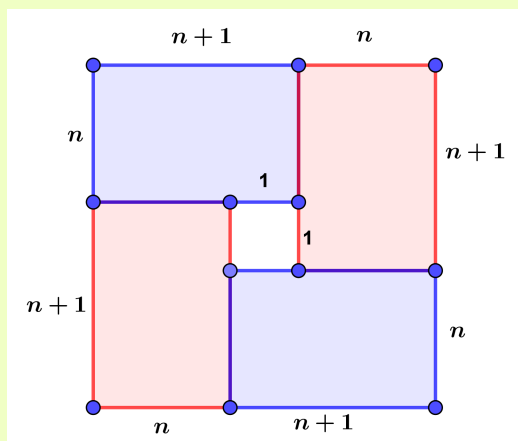


Figure 14: Visual demonstration of the identity $8T_n + 1 = (2n + 1)^2$

In the central space there is an uncovered 1×1 square. We see that $4 \times n(n + 1) + 1 = (2n + 1)^2$ and hence that $8T_n + 1 = (2n + 1)^2$.

More connections with the square numbers In addition to the above, there are many identities that are more challenging to interpret visually (or combinatorially). But they are easy to verify using algebra. We shall not give the algebraic details (please do this for yourself).

We invite the reader to look for visual (or any other) interpretations of these identities.

$$1. \quad T_n + T_{n+3} = (n+2)^2 + 2$$

$$\text{Example: } T_2 + T_5 = 3 + 15 = 18 = 4^2 + 2$$

$$2. \quad T_n + T_{n+5} = (n+3)^2 + 6$$

$$\text{Example: } T_3 + T_8 = 6 + 36 = 42 = 6^2 + 6$$

$$3. \quad T_n + T_{n+7} = (n+4)^2 + 12$$

$$\text{Example: } T_3 + T_{10} = 6 + 55 = 61 = 7^2 + 12$$

$$4. \quad T_n + T_{n+2k-1} = (n+k)^2 + 2T_{k-1}$$

As can be seen, there are infinitely many such identities.

Identities involving triples of triangular numbers Here is another infinite sequence of identities.

$$1. \quad T_n + T_{n+2} = 2T_{n+1} + 1^2$$

$$\text{Example: } T_2 + T_4 = 3 + 10 = 13 = 2T_3 + 1$$

$$2. \quad T_n + T_{n+4} = 2T_{n+2} + 2^2$$

$$\text{Example: } T_3 + T_7 = 6 + 28 = 34 = 2T_5 + 4$$

$$3. \quad T_n + T_{n+6} = 2T_{n+3} + 3^2$$

$$\text{Example: } T_3 + T_9 = 6 + 45 = 51 = 2T_6 + 9$$

$$4. \quad T_n + T_{n+2k} = 2T_{n+k} + k^2$$

These too are challenging to interpret visually (or combinatorially). But they are easy to verify using algebra. We invite the reader to look for visual (or any other) interpretations of these identities.

How do you recognize the triangular numbers?

Given an arbitrary positive integer, is there a simple way of checking whether it is a triangular number?

To explain the context of this question, consider the corresponding question for the square numbers: “Given an arbitrary positive integer, is there a simple way of checking whether it is a square number?” Yes, there is: just compute the square root of the number!

In the same spirit we ask: “Given an arbitrary $n \in \mathbb{N}$, how shall we check whether there exists $k \in \mathbb{N}$ such that $T_k = n$?” The question turns out to have an easy answer.

We proved earlier that $8T_n + 1 = (2n + 1)^2$. This implies the following:

If n is a triangular number, then $8n + 1$ is a square number.

We shall show that the implication goes in the reverse direction too:

If n is such that $8n + 1$ is a square number, then n is a triangular number.

Proof: Assume that $n \in \mathbb{N}$ is such that $8n + 1$ is a square number. Since $8n + 1$ is odd, this means that

$$8n + 1 = (2a + 1)^2,$$

for some $a \in \mathbb{N}$. This yields

$$8n + 1 = 4a^2 + 4a + 1,$$

$$\therefore n = \frac{a^2 + a}{2} = \frac{a(a + 1)}{2},$$

i.e., $n = T_a$. This justifies the claim. ■

A result of Euler's

We close this article with a description of a beautiful result proved by Euler.

Theorem (Euler). *If $n \in \mathbb{N}$ is a triangular number, then the following are also triangular numbers: $9n + 1$, $25n + 3$, and $49n + 6$.*

Proof: Let $n \in \mathbb{N}$ be a triangular number, say $n = k(k+1)/2$. We must show that $9n + 1$ is a triangular number. We make use of the criterion proved above. This means that we must prove that $8(9n + 1) + 1$ is a square number. We have now,

$$\begin{aligned} 8(9n + 1) + 1 &= 8 \left(\frac{9k(k+1)}{2} + 1 \right) + 1 \\ &= 36k^2 + 36k + 9 = (6k + 3)^2. \end{aligned}$$

We see that if n is a triangular number, then $8(9n + 1) + 1$ is a square number, implying that $9n + 1$ is a triangular number.

Similarly, in the case of $25n + 3$, we must prove that $8(25n + 3) + 1$ is a square number. We have now, with $n = k(k+1)/2$,

$$\begin{aligned} 8(25n + 3) + 1 &= 8 \left(\frac{25k(k+1)}{2} + 3 \right) + 1 \\ &= 100k^2 + 100k + 25 = (10k + 5)^2, \end{aligned}$$

hence $25n + 3$ is a triangular number.

We leave the verification of the third case to the reader.

Generalization of Euler's result

Let us now tackle a more challenging problem. Namely, let us see if we can express Euler's finding in a more general form.

We shall explore the following problem.

Problem: Find all $a, b \in \mathbb{N}$ such that if $n \in \mathbb{N}$ is a triangular number, then so is $an + b$.

Analysis: Assume that $n \in \mathbb{N}$ is a triangular number. Then $n = k(k+1)/2$ for some $k \in \mathbb{N}$.

So, we must solve the following quadratic equation for m ,

$$\frac{m(m+1)}{2} = a \cdot \frac{k(k+1)}{2} + b,$$

and find the conditions on a, b under which m is an integer for all integer values of k .

Rewriting the equation as

$$m^2 + m - ak(k+1) - 2b = 0,$$

we obtain the roots:

$$m = \frac{-1 \pm \sqrt{4ak(k+1) + 8b + 1}}{2}.$$

As m is an integer, the quantity under the square root sign, namely:

$$4ak^2 + 4ak + 8b + 1,$$

must be a square number for all integer values of k . But this can only be the case if the expression $4ak^2 + 4ak + 8b + 1$ is algebraically a perfect square.

This last condition implies the following.

- $8b + 1$ is a square number, say $8b + 1 = c^2$. This means that b is a triangular number. Note that c is odd.
- a is a square number, say $a = d^2$.

Making the relevant substitutions, we obtain the expression

$$4d^2k^2 + 4d^2k + c^2.$$

Equating this to $(2dk + c)^2$, we get

$$4cdk = 4d^2k, \quad \therefore c = d.$$

Our findings yield a complete answer to the problem. We express it in the form of a recipe as follows.

- Choose b to be any triangular number. Let $a = 8b + 1$.
- Then the expression $an + b$ satisfies the stated condition. That is, if n is a triangular number, then $an + b$ too is a triangular number.
- Moreover, this recipe yields all possible expressions $an + b$ satisfying the stated condition.

Example: Choose $b = 1$; then $a = 9$. This yields the expression $9n + 1$.

Or choose $b = 3$; then $a = 25$. This yields the expression $25n + 3$.

Or choose $b = 6$; then $a = 49$. This yields the expression $49n + 6$.

Etc. I'm sure the reader will agree that this is a beautiful result.

Article contributed by the editors

Sums of Cubes – 1

Most of us have heard the taxicab story [1], [2] featuring G H Hardy and S Ramanujan and the number 1729. Here it is, in Hardy's own words:

I remember once going to see him when he was ill at Putney. I had ridden in taxi cab No. 1729 and remarked that the number seemed to me rather a dull one, and that I hoped it was not an unfavourable omen. "No," he replied, "it is a very interesting number; it is the smallest number expressible as the sum of two cubes in two different ways."

The numerical relation that Ramanujan had in mind was this:

$$1729 = 9^3 + 10^3 = 1^3 + 12^3.$$

As a result of this episode, numbers with the property that they can be expressed as the sum of two positive cubes in more than one way are now known as *taxicab numbers*. Many questions can be asked about these numbers. We shall explore some of them in this article.

Note that there is another way in which we can look at the above equality:

$$9^3 + 10^3 = 12^3 + 1.$$

This means that the triple $(9, 10, 12)$ forms a “near miss” to the cubic case of Fermat’s equation, $x^3 + y^3 = z^3$. It is well known that there are no positive integer triples that satisfy this equation. So, at best, we can ask for “near misses” — i.e., positive integer triples (x, y, z) for which

$$x^3 + y^3 = z^3 \pm 1.$$

We will examine this equation in a future ‘episode’ of this series.

Sums of two cubes

Which positive integers are sums of two cubes? To start with, we permit the use of cubes of negative integers. So the question is:

Which positive integers n can be written in the form $n = a^3 + b^3$ where $a, b \in \mathbb{Z}$?

There are (obviously) many more numbers which are sums of two cubes than numbers which are cubes. How many more? What can be said about these numbers? Let S represent the set of all positive integers n which can be written in this form:

$$S = \left\{ n : n = a^3 + b^3, a \in \mathbb{Z}, b \in \mathbb{Z} \right\}.$$

We display below the first 100 numbers in S (note that S contains all the cubes).

1	2	7	8	9	16	19	26	27	28
35	37	54	56	61	63	64	65	72	91
98	117	124	125	126	127	128	133	152	169
189	208	215	216	217	218	224	243	250	271
279	280	296	316	331	335	341	342	343	344
351	370	386	387	397	407	432	448	468	469
485	488	504	511	512	513	520	539	547	559
576	602	604	631	637	657	665	686	702	721
728	729	730	737	756	784	793	817	819	854
855	866	875	919	936	945	973	988	992	999

Table 1: Sums of two cubes (with negative cubes permitted)

Inputs from modular arithmetic

In exploring the structure of a set of integers generated through any arithmetical procedure, it often helps to examine the set through the lens of modular arithmetic. We shall do the same with the set S .

Consider the possible remainders left when the cubes are divided by various natural numbers. On division by 2, remainders of 0 and 1 are possible, and on division by 3, remainders of 0, 1 and 2 are possible; no surprises in either of these. On division by 4, remainders of 0, 1 and 3 are possible, but not a remainder of 2 (please check).

Continuing, we obtain the result shown in the table below.

Modulus	Remainders	Non-remainders
2	0, 1	
3	0, 1, 2	
4	0, 1, 3	2
5	0, 1, 2, 3, 4	
6	0, 1, 2, 3, 4, 5	
7	0, 1, 6	2, 3, 4, 5
8	0, 1, 3, 5, 7	2, 4, 6
9	0, 1, 8	2, 3, 4, 5, 6, 7

Table 2

We see that the first interesting cases are when the moduli are 7 and 9, as there are more non-remainders than remainders in these cases. This enables the possibility of using these two moduli for characterising numbers in S .

In the analysis below, we use only the modulus 9. The fact that the only remainders possible are 0, 1, 8 allows us to state the following.

Theorem 1. *Every cube is of one of the following forms: $9k$, $9k \pm 1$.*

This immediately implies the following two corollaries (the second one is

the first one stated in contrapositive form):

Corollary 1. *Every number in S is of one of these forms: $9k$, $9k \pm 1$, $9k \pm 2$.*

Corollary 2. *If a number is of the form $9k \pm 3$ or $9k \pm 4$, it does not belong to S .*

The condition in the corollary does not eliminate sufficiently many numbers from membership in S . We need better characterisations of S , but these are not readily forthcoming.

We may contrast this with the situation for squares, when we have a very compact characterisation available: *A number n is expressible as a sum of two squares if and only if, in the prime factorisation of n , all prime factors of the form $4k + 3$ occur with even exponent.*

For the sum-of-two cubes problem, though characterisations are available, they are quite involved. For a recent result in this area, see [4].

The prime numbers in S

The prime numbers among the first 100 numbers in S are the following:

2, 7, 19, 37, 61, 127, 271, 331, 397, 547, 631, 919.

If we discard the prime number 2, then a curious pattern is noticed about all the remaining numbers: *they are all of the form $1 \pmod{9}$ or $-2 \pmod{9}$* . Is this a genuine pattern, i.e., true for all the odd primes in S , or a misleading one that persists only among the first few numbers in S (and lets us down after that)? If true, it would imply that numbers in S which exceed 2 and are of the form $-1 \pmod{9}$ or $2 \pmod{9}$ are all composite. So is this pattern genuine? The answer is not known.

Back to taxicab numbers

Returning to the study of taxicab numbers, we now reimpose the restriction of using only cubes of positive integers. So our search is for numbers with the property that *they can be expressed as the sum of two positive cubes in more than one way*. This means that we need an effective way of finding solutions to the following Diophantine equation:

$$x^3 + y^3 = u^3 + v^3, \quad \text{with } x, y, u, v \in \mathbb{N}.$$

Thanks to Hardy's colourful telling of the taxicab tale, this equation will for ever be associated with Ramanujan. But back in the 18th century, Euler had studied the very same equation and had found a complete parametric solution. Ramanujan too found a parametric solution, and that is the one we will examine now. We must mention here that though Ramanujan's solution yields infinitely many solutions to the above equation (and therefore infinitely many taxicab numbers), it does not yield all of them. But it is easier to work with, hence our choice.

Ramanujan cleverly uses the following well known equality,

$$3^3 + 4^3 + 5^3 = 6^3$$

(which is true, though it may seem hard to believe; please check!) to concoct the following algebraic identity:

$$(3t^2 + 5t - 5)^3 + (4t^2 - 4t + 6)^3 + (5t^2 - 5t - 3)^3 = (6t^2 - 4t + 4)^3.$$

We invite you to verify the identity on your own.

Remark: Note carefully the connection between the identity and the numerical equality $3^3 + 4^3 + 5^3 = 6^3$. For example, look at the coefficients

of the leading terms both sides. On the LHS we get $3^3 + 4^3 + 5^3$; this will be the coefficient of t^6 . And on the RHS we get 6^3 . These two quantities are equal, as noted above.

Similarly, look at the constant terms on both sides. On the LHS we get $(-5)^3 + 6^3 + (-3)^2$, and on the RHS we get 4^3 . The two quantities are equal.

Fixing the coefficients of t in the four polynomials so that the coefficients of t^5, t^4, t^3, t^2, t come out as desired would have required careful juggling of the various possibilities. But Ramanujan managed this (with ease, we are sure).

Using the parametric formula to generate taxicab numbers: We can now give to t any rational value that we please, other than 1. The parametric relation listed above then generates an instance of a taxicab number.

Example 1: Let $t = 2$. Then we get

$$3t^2 + 5t - 5 = 17, \qquad 4t^2 - 4t + 6 = 14,$$

$$5t^2 - 5t - 3 = -7, \qquad 6t^2 - 4t + 4 = 20.$$

So we get

$$17^3 + 14^3 = (-7)^3 + 20^3.$$

A nicer way of writing this is

$$17^3 + 14^3 + 7^3 = 20^3.$$

Note that we cannot call 7657 a taxicab number because of the presence of a negative integer on the right side.

Example 2: Let $t = 3$. Then we get

$$3t^2 + 5t - 5 = 37,$$

$$4t^2 - 4t + 6 = 30,$$

$$5t^2 - 5t - 3 = -27,$$

$$6t^2 - 4t + 4 = 46.$$

So we get

$$37^3 + 30^3 + 27^3 = 46^3.$$

Example 3: Let $t = 4$. Then we get

$$3t^2 + 5t - 5 = 63,$$

$$4t^2 - 4t + 6 = 54,$$

$$5t^2 - 5t - 3 = -57,$$

$$6t^2 - 4t + 4 = 84.$$

So we get

$$63^3 + 54^3 + 57^3 = 84^3.$$

Fixing the negative numbers: The examples above have not yielded any taxicab numbers because of the presence of negative numbers. So, we must find a way of “fixing” the negative numbers. We first identify the range of values of t for which

$$3t^2 + 5t - 5 > 0,$$

$$4t^2 - 4t + 6 > 0,$$

$$5t^2 - 5t - 3 < 0,$$

$$6t^2 - 4t + 4 > 0.$$

Each inequality is easily solved. The intersection of the four solution sets turns out to be the following interval:

$$\frac{\sqrt{85} - 5}{6} < t < \frac{\sqrt{85} + 5}{10},$$

or, approximately,

$$0.7033 < t < 1.4219.$$

So, we must select rational values of t only within this interval. The values used earlier ($t = 2, 3, 4$) all lie outside this interval.

Example 4: Let $t = 4/3$ (this lies within the above interval). We get

$$\begin{aligned} 3t^2 + 5t - 5 &= 7, & 4t^2 - 4t + 6 &= \frac{70}{9}, \\ 5t^2 - 5t - 3 &= \frac{7}{9}, & 6t^2 - 4t + 4 &= \frac{28}{3}. \end{aligned}$$

This yields the equality

$$7^3 + \left(\frac{70}{9}\right)^3 = \left(\frac{7}{9}\right)^3 + \left(\frac{28}{3}\right)^3.$$

Clearing the fractions, this is equivalent to:

$$63^3 + 70^3 = 7^3 + 84^3.$$

All four numbers in the above relation are multiples of 7. Dividing out by this common factor, we are left with the following:

$$9^3 + 10^3 = 1^3 + 12^3.$$

Well, we have stumbled upon Ramanujan's taxicab number!

Example 5: Let $t = 5/4$ (this too lies within the above interval). We get

$$\begin{aligned} 3t^2 + 5t - 5 &= \frac{95}{16}, & 4t^2 - 4t + 6 &= \frac{29}{4}, \\ 5t^2 - 5t - 3 &= \frac{23}{16}, & 6t^2 - 4t + 4 &= \frac{67}{8}. \end{aligned}$$

This yields the equality

$$\left(\frac{95}{16}\right)^3 + \left(\frac{29}{4}\right)^3 = \left(\frac{23}{16}\right)^3 + \left(\frac{67}{8}\right)^3.$$

Clearing the fractions, this is equivalent to:

$$95^3 + 116^3 = 23^3 + 134^3.$$

Both sides are equal to 2418271. By definition, this is a taxicab number. It is substantially larger than the Ramanujan taxicab number.

Proceeding in this manner, we may compute as many taxicab numbers as we wish.

List of taxicab numbers: We invite the reader to refer to [3], which gives a list of taxicab numbers. Here are the first thirty of these numbers.

1729	4104	13832	20683	32832
39312	40033	46683	64232	65728
110656	110808	134379	149389	165464
171288	195841	216027	216125	262656
314496	320264	327763	373464	402597
439101	443889	513000	513856	515375

Table 3

The second number on the list, 4104, is noteworthy as it admits *three* different representations:

$$4104 = 16^3 + 2^3 = 15^3 + 9^3 = (-12)^3 + 18^3.$$

Article contributed by Shailesh Shirali

Further reading

- [1] Wikipedia, “Taxicab number” from
https://en.wikipedia.org/wiki/Taxicab_number
- [2] Wikipedia, “1729” from
[https://en.wikipedia.org/wiki/1729_\(number\)](https://en.wikipedia.org/wiki/1729_(number))
- [3] The On-Line Encyclopedia of Integer Sequences, sequence A001235,
from <https://oeis.org/A001235>
- [4] Kevin A. Broughan, “Characterizing the Sum of Two Cubes”, Journal
of Integer Sequences, Vol. 6 (2003), [https://cs.uwaterloo.ca/
journals/JIS/VOL6/Broughan/broughan25.pdf](https://cs.uwaterloo.ca/journals/JIS/VOL6/Broughan/broughan25.pdf)

Fascinating Numbers – 1

Article contributed by the editors

Two articles in this issue deal with particular families of numbers — *Triangular Numbers* and *Taxicab Numbers*. In this article we list without proof more properties of triangular numbers. Much of the material is from [1], a book that we highly recommend to all number enthusiasts.

Some fascinating features of triangular numbers

1. “Flocks of birds often fly in this triangular formation. Even several airplanes flying together constitute this formation.” (Page 84, Chapter 3 of [1])
2. From the sequence of triangular numbers $(1, 3, 6, 10, \dots)$, let us subtract, term by term, the odd numbers $(1, 3, 5, 7, \dots)$. If we discard the first two terms of the resulting sequence, we get back the sequence of triangular numbers (Page 85, Chapter 3 of [1]):

0, 0, 1, 3, 6, 10, ...

3. The sum of the squares of two consecutive triangular numbers is also a triangular number (Page 86, Chapter 3 of [1]):

$$(T_{n-1})^2 + (T_n)^2 = T_{n^2}.$$

4. Twice the n -th triangular number plus the square of n gives the $2n$ -triangular number (Page 86, Chapter 3 of [1]):

$$2T_n + n^2 = T_{2n}.$$

An elegant visual proof of this identity is given on the same page.

5. The units digit of a triangular number can only be 0, 1, 3, 5, 6 or 8. It can never be any of the digits 2, 4, 7 or 9 (Page 86, Chapter 3 of [1]).
6. They do not exist four distinct triangular numbers in arithmetic progression (AP). However, there are infinitely many triples of distinct triangular numbers in AP (Page 88, Chapter 3 of [1]). Examples of the latter: (1,28,55), (3,78,153), (6,36,66), (10,253,496), ...
7. There do not exist four distinct triangular numbers in geometric progression (GP). However, there are infinitely many triples of distinct triangular numbers in GP (Page 90, Chapter 3 of [1]). Examples of the latter: (1,6,36), (36,210,1225), (1225,7140,41616), ...
8. There are infinitely many triangular numbers that are the products of two consecutive numbers (Page 91, Chapter 3 of [1]). For example:

$$T_3 = 6 = 2 \times 3,$$

$$T_{20} = 210 = 14 \times 15,$$

$$T_{119} = 7140 = 84 \times 85,$$

$$T_{696} = 242556 = 492 \times 493.$$

9. There are some triangular numbers that are the products of three consecutive numbers (Page 91-92, Chapter 3 of [1]). For example:

$$T_3 = 6 = 1 \times 2 \times 3,$$

$$T_{15} = 120 = 4 \times 5 \times 6,$$

$$T_{20} = 210 = 5 \times 6 \times 7,$$

etc. It is not clear if this list can be extended indefinitely.

10. The triangular number $T_{15} = 120$ has the unusual feature that it is the product of three, four and five consecutive numbers (Page 92, Chapter 3 of [1]):

$$T_{15} = 120 = 4 \times 5 \times 6 = 2 \times 3 \times 4 \times 5 = 1 \times 2 \times 3 \times 4 \times 5.$$

It is not known whether there are any other triangular numbers with such a feature.

We have not provided proofs of any of the assertions. We invite readers to supply them!

Further reading

- [1] Shyam Sunder Gupta, "Exploring the beauty of fascinating numbers"
Springer

Part III

Contributed Articles – 1

Fast Calculations – 3

Article contributed by S. C. Dutta Roy

We continue our study of computational shortcuts (or, “the art of doing fast calculations and impressing an audience”).

In the article [1] “Fast Calculations – 2”, published in *The Compass*, Vol. 1-2, pp. 1-2, October 2024, a short-cut was given for finding the square of any two-digit number whose first digit is 1 or 2. The article ended with asking the reader to find similar short-cuts for finding the square of a two-digit number whose first digit is 3 or 4. We give here such a short-cut that works for any two-digit number and give examples to illustrate it.

Short-cut for squaring a two-digit number

Let the number be $\overline{ab} = 10a + b$. Here are the steps to find $(\overline{ab})^2$:

1. Add b to $\overline{ab}$
2. Multiply the result by $10a$
3. Add b^2 to the result

Example 1

Let us find the value of 56^2 . Following the above steps, we obtain:

1. $56 + 6 = 62$
2. $62 \times 50 = 3100$
3. $3100 + 6^2 = 3136$. This is the required answer.

Example 2

Let us find the value of 63^2 . Following the above steps, we obtain:

1. $63 + 3 = 66$
2. $66 \times 60 = 3960$
3. $3960 + 3^2 = 3969$. This is the required answer.

Explanation

The logic for the procedure can be worked out in the same manner as in [1], but we use here a slightly different approach. We have,

$$\overline{ab}^2 = (10a + b)^2 = 100a^2 + b^2 + 20ab.$$

The steps described in the procedure yield:

$$\begin{aligned}\overline{ab} &= 10a + b \mapsto \overline{ab} + b = 10a + 2b \\ &\mapsto 10a \times (10a + 2b) = 100a^2 + 20ab \\ &\mapsto 100a^2 + 20ab + b^2.\end{aligned}$$

Comment

It is not easy to formulate a short-cut for finding the square of a three-digit number or the cube of a two-digit number. On the other hand, it is easy to apply the straightforward algebraic procedure.

Concluding remarks

One might wonder why in this age of calculators, smartphones and laptops, one should use short-cuts for finding the squares and cubes of numbers.

The author sees two reasons. First, to generate interest in students, and second, for their use during examinations, when these implements are not available.

About the author

Prof Dutta Roy spent about four decades in the Electrical Engineering Department of Electrical Engineering, Indian Institute of Technology Delhi, and is now retired. Currently, his interests are in exploring and appreciating the beauty of Classical Music and Mathematics. He may be contacted at s.c.dutta.roy@gmail.com.

The Inescapable Friday the 13th

Article contributed by Prithwiji De

In any Gregorian Calendar the ominous [Friday the 13th](#) will always occur, and there can be at most three of them.

Why? To see this consider a 365-day year and number the 13th day of every month by the number of the day on which it falls. So, January 13 is assigned the number 13, February 13 is assigned $31 + 13 = 44$, and so on. Continuing, we obtain the following list of 12 numbers.

13, 44, 72, 103, 133, 164, 194, 225, 256, 286, 317, 347.

Without further information, it is not clear if any of these days correspond to a Friday of some week of the year. But do we really need to know what's the first day of the year to figure out if a Friday has appeared in the list above? Knowing the first day would no doubt tell us the answer immediately, but we can do without it.

Let us assign these numbers to the days of a week. The 13-th day of the year is the sixth day of the second week. Note that it may not be a Saturday because we don't know the first day of the year. Just to clarify, a week here corresponds to a block of seven successive days and each day is assigned a number from the set $\{0, 1, 2, 3, 4, 5, 6\}$. So we assign the number 6 to the 13-th day.

Next, the 44-th day of the year is assigned the number 2 because it is the second day of the seventh block from the beginning. Proceeding similarly we end up with the 12 numbers

$$6, 2, 2, 5, 0, 3, 5, 1, 4, 6, 2, 4$$

and we observe that every member of the set $\{0, 1, 2, 3, 4, 5, 6\}$ has appeared in the above list, and any member of the set $\{0, 1, 2, 3, 4, 5, 6\}$ has appeared *at most thrice*.

The moral of the story is that in any given year, the 13-th of a month can fall on any day of the week, and it can fall on the same day of the week at most thrice. Thus, there is no escaping the *Friday the 13th* in a year, and one may have to face it not just once or twice but thrice.

What happens in a leap year? It is the same story.

Can any particular day of the week be avoided?

One may wonder if there is any date of a month that always avoids a certain day of a week.

There isn't such a date. To see this observe that the k -th day of a month

corresponds to one of the days in the list

$$k, k + 31, k + 59, k + 90, k + 120, k + 151,$$

$$k + 181, k + 212, k + 243, k + 273, k + 304, k + 334.$$

If the numbers in this list are divided by 7, the remainders obtained are

$$a, a + 3, a + 3, a + 6, a + 1, a + 4,$$

$$a + 6, a + 2, a + 5, a, a + 3, a + 5,$$

where a is the remainder obtained upon dividing k by 7. Subtracting a from every member of the list we are left with

$$0, 3, 3, 6, 1, 4, 6, 2, 5, 0, 3, 5$$

which shows that all days of a week are represented. Thus there isn't a date in a 365-day year which avoids a particular day of the week.

About the author

Prithwijit De is the National Coordinator of the Mathematical Olympiad Programme of the Government of India. He is an Associate Professor at the Homi Bhabha Centre for Science Education (HBCSE), TIFR, Mumbai. He loves to read and write popular articles in mathematics as much as he enjoys mathematical problem solving.

He may be contacted at de.prithwijit@gmail.com

Summing the Cubes of the Natural Numbers – I

Article contributed by Saibal Gupta

Introduction: The technique of summing the natural numbers from 1 to n using “reversal of terms” (the method generally attributed to Carl Gauss) is well known. (*Note from the editor.* For more on this episode, please see the article on triangular numbers elsewhere in this issue of *The Compass*.) Would this technique work for other sequences? Here we show that it is possible to use this idea to find the sum of the cubes of the natural numbers from 1 to n . The details are given below.

Analysis: The basic algebraic identity we use repeatedly is:

$$a^3 + b^3 = (a + b) \cdot (a^2 - ab + b^2).$$

Let S denote the sum of the cubes of the natural numbers from 1 to n :

$$S = 1^3 + 2^3 + 3^3 + \cdots + n^3.$$

We reverse the order of the terms in the sum and write both the sums, lining up the terms vertically. Then S is equal to both of the following:

$$\begin{array}{cccccccc} 1^3 & + & 2^3 & + & 3^3 & + & \cdots & + & (n-1)^3 & + & n^3 \\ n^3 & + & (n-1)^3 & + & (n-2)^3 & + & \cdots & + & 2^3 & + & 1^3 \end{array}$$

Pairing up the terms “vertically” and adding, as in the Gauss story, we get

$$2S = (n^3 + 1^3) + ((n-1)^3 + 2^3) + \cdots + (1^3 + n^3).$$

Using the factor identity mentioned above, we see that each bracketed term has $n+1$ as a factor. Hence:

$$2S = (n+1) \cdot A,$$

where

$$A = (n^2 - n + 1^2) + ((n-1)^2 - 2(n-1) + 2^2) + \cdots + (1^2 - n + n^2).$$

The general term of the expression on the right side simplifies to

$$\begin{aligned} & (n-a+1)^2 - a(n-a+1) + a^2 \\ &= n^2 + n(2-3a) + 3a^2 - 3a + 1 \\ &= n^2 - (3a-2)n + (a^3 - (a-1)^3), \end{aligned}$$

where a takes the values $1, 2, \dots, n$.

Now let us sum these expressions for $a = 1, 2, \dots, n$.

Here is what we get.

- There are n repetitions of n^2 , which yield n^3 on summation.

- The sum of the linear expressions in n is

$$\begin{aligned}\sum_{a=1}^n (3a - 2)n &= (3 \cdot 1 - 2)n + (3 \cdot 2 - 2)n + \cdots + (3 \cdot n - 2)n \\ &= 3(1 + 2 + 3 + \cdots + n) \cdot n - 2n \cdot n \\ &= \frac{3n^2(n+1)}{2} - 2n^2 = \frac{n^2(3n-1)}{2}.\end{aligned}$$

- For the last two (cubic) terms, a nice telescopic simplification occurs:

$$(1^3 - 0^3) + (2^3 - 1^3) + (3^3 - 2^3) + \cdots + (n^3 - (n-1)^3) = n^3.$$

Hence:

$$A = n^3 - \frac{n^2(3n-1)}{2} + n^3 = \frac{n^2(n+1)}{2},$$

on simplification. Since $2S = (n+1)A$, it follows that

$$S = \frac{n^2(n+1)^2}{4}.$$

Corollary

We already know that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

This means that we have the remarkable equality

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = (1 + 2 + 3 + \cdots + n)^2.$$

Note from the editor: The article immediately following this gives a very different proof of the same identity.

About the author

Saibal Gupta did his B-Tech and M-Tech in Radio Physics & Electronics, CU. Thereafter, he served the DoT and BSNL in various capacities. He retired from BSNL as Senior GM. Subsequently, he served as a Guest faculty in a private Engineering College, ECE Dept. He is a Fellow of IETE. He may be contacted at saibalgupta61@gmail.com.

Summing the Cubes of the Natural Numbers – II

Article contributed by S. C. Dutta Roy

We offer another derivation here of the formula for the sum of the cubes of the first n natural numbers. For positive integers n, k , define

$$S_{n,k} := 1^k + 2^k + 3^k + \cdots + n^k.$$

We already know that

$$S_{n,1} = \frac{n(n+1)}{2}.$$

Let us prove the following identity:

$$S_{n,3} = S_{n,1}^2.$$

Proof: By definition of $S_{n,k}$ we have,

$$S_{n+1,3} - S_{n,3} = (n+1)^3$$

$$S_{n+1,1} - S_{n,1} = n+1.$$

We also have, from the identity $(n+1)^2 = n(n+1) + (n+1)$,

$$(n+1)^2 = 2 \cdot S_{n,1} + (n+1).$$

Hence we can write

$$\begin{aligned} S_{n+1,3} - S_{n,3} &= (n+1) \cdot (n+1)^2 \\ &= (S_{n+1,1} - S_{n,1}) \cdot (2 \cdot S_{n,1} + (n+1)) \\ &= (S_{n+1,1} - S_{n,1}) \cdot (2 \cdot S_{n,1} + S_{n+1,1} - S_{n,1}) \\ &= (S_{n+1,1} - S_{n,1}) \cdot (S_{n+1,1} + S_{n,1}) \\ &= S_{n+1,1}^2 - S_{n,1}^2. \end{aligned}$$

Hence $S_{n+1,3} - S_{n+1,1}^2 = S_{n,3} - S_{n,1}^2$. From this we conclude that

$$S_{n+1,3} - S_{n+1,1}^2 = S_{n,3} - S_{n,1}^2 = S_{n-1,3} - S_{n-1,1}^2 = \cdots = S_{1,3} - S_{1,1}^2.$$

Since $S_{1,3} - S_{1,1}^2 = 0$, trivially, it follows that

$$S_{n,3} = S_{n,1}^2 \quad \text{for all } n \in \mathbb{N}.$$

About the author

Prof Dutta Roy spent about four decades in the Electrical Engineering Department of Electrical Engineering, Indian Institute of Technology Delhi, and is now retired. Currently, his interests are in exploring and appreciating the beauty of Classical Music and Mathematics. He may be contacted at s.c.dutta.roy@gmail.com.

A Conditional Trigonometric Identity

Article contributed by Akash Upadhyay

In this short article, we study the trigonometric expression

$$E = \tan(A + B) + \cot(A - B),$$

and derive a conditional identity.

Let A, B be angles. Then we have the identities

$$\tan(A + B) = \frac{\sin(A + B)}{\cos(A + B)},$$

$$\cot(A - B) = \frac{\cos(A - B)}{\sin(A - B)},$$

which hold provided $\cos(A + B) \neq 0$ and $\sin(A - B) \neq 0$.

Hence, given these two conditions,

$$\begin{aligned} E &= \frac{\sin(A+B)}{\cos(A+B)} + \frac{\cos(A-B)}{\sin(A-B)} \\ &= \frac{\sin(A+B)\sin(A-B) + \cos(A+B)\cos(A-B)}{\cos(A+B)\sin(A-B)}. \end{aligned}$$

Using the trigonometric identity

$$\cos X \cos Y + \sin X \sin Y = \cos(X - Y)$$

with $X = A + B$ and $Y = A - B$, we obtain

$$E = \frac{\cos[(A+B) - (A-B)]}{\cos(A+B)\sin(A-B)},$$

i.e.,

$$E = \frac{\cos 2B}{\cos(A+B)\sin(A-B)}.$$

Hence we have,

$$\tan(A+B) + \cot(A-B) = \frac{\cos(2B)}{\cos(A+B)\sin(A-B)},$$

provided $\cos(A+B) \neq 0$ and $\sin(A-B) \neq 0$.

Remarks

The above identity is not universally true. **Rather, it is conditionally valid.**
The conditions for validity are as noted above.

Such relations can prove useful for teaching purposes to demonstrate the relevance and importance of domain restrictions.

About the author

Akash Upadhyay is a mathematics educator with experience in teaching different curricula. He holds a Master's degree in Mathematics and is passionate about making the subject intuitive and engaging for students. He is the author of *Making Mathematics Come Alive: A Teacher's Guide to Wonder, Rhythm, Story, and Practice*. His work focuses on blending conceptual clarity with creative pedagogy to enhance learning experiences. He may be contacted at akashupadhyayoms555@gmail.com.

Areas of Regions Defined by Circles

Article contributed by Triparna Das

Introduction: The problem of finding areas formed by intersecting circles is a classical topic in plane geometry. In this work, we derive a general expression for the area of the curvilinear triangle formed by three mutually tangent circles (tangent externally). We use the Law of Cosines, Heron's formula and simple geometry. The area is expressed in terms of the radii r_i and the angles θ_i subtended at the centres (in radian measure).

Computing the area: Let three circles centred at A, B, C be externally tangent to each other at X, Y, Z . Let their radii be r_1, r_2, r_3 respectively. Let the angles subtended by arcs XZ, XY, YZ be $\theta_1, \theta_2, \theta_3$ respectively. Our aim is to find the area of the curvilinear triangle XYZ (Figure 15).

We have,

$$AB = r_1 + r_2 = c, \quad BC = r_2 + r_3 = a, \quad CA = r_3 + r_1 = b.$$

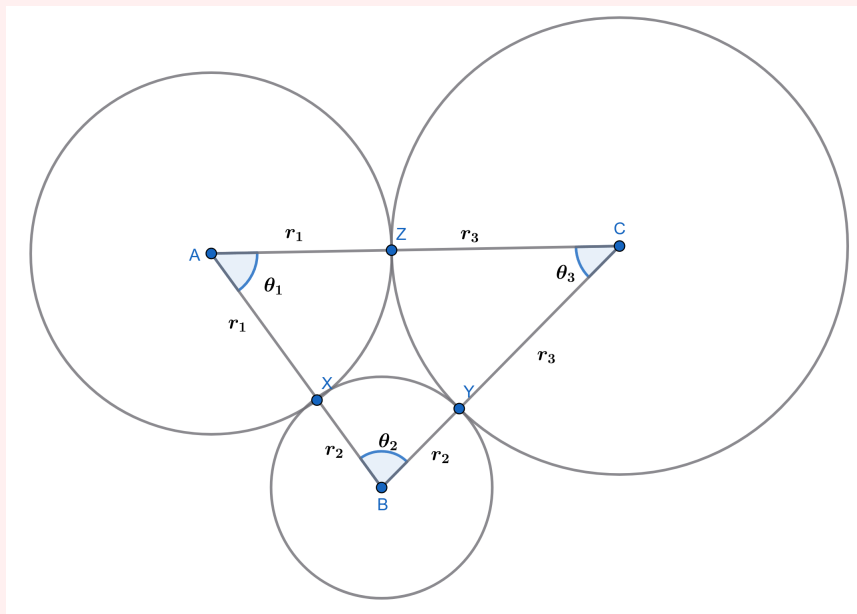


Figure 15

Using Heron's Formula: The semi-perimeter s of triangle $\triangle ABC$ is

$$s = \frac{AB + BC + CA}{2} = \frac{(r_1 + r_2) + (r_2 + r_3) + (r_3 + r_1)}{2}$$

$$= r_1 + r_2 + r_3.$$

Hence the area P of ABC is

$$P = \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \sqrt{r_1 r_2 r_3 (r_1 + r_2 + r_3)}.$$

Areas of the sectors: The area of the sector of the circle of radius r_i contained within $\triangle ABC$ is $\frac{1}{2}r_i^2\theta_i$. Hence the total area Q of the three sectors is

$$Q = \frac{1}{2} \left(r_1^2\theta_1 + r_2^2\theta_2 + r_3^2\theta_3 \right).$$

Area of the curvilinear triangle: It follows that the area $K = P - Q$ of the curvilinear triangle XYZ is given by

$$K = \sqrt{r_1 r_2 r_3 (r_1 + r_2 + r_3)} - \frac{1}{2} \left(r_1^2\theta_1 + r_2^2\theta_2 + r_3^2\theta_3 \right).$$

Hence:

Area Of Curvilinear Triangle

$$K = \sqrt{\left(\prod_{i=1}^3 r_i \right) \left(\sum_{i=1}^3 r_i \right)} - \frac{1}{2} \sum_{i=1}^3 r_i^2 \theta_i.$$

Using the law of cosines: By using the cosine rule, we may rewrite the above formula.

We obtain the following:

$$K = \sqrt{\left(\prod_{i=1}^3 r_i\right) \left(\sum_{i=1}^3 r_i\right)} \\ - \frac{1}{2} r_1^2 \cos^{-1} \left(\frac{b^2 + c^2 - a^2}{2bc} \right) \\ - \frac{1}{2} r_2^2 \cos^{-1} \left(\frac{a^2 + c^2 - b^2}{2ac} \right) \\ - \frac{1}{2} r_3^2 \cos^{-1} \left(\frac{a^2 + b^2 - c^2}{2ab} \right).$$

By substituting $a = r_2 + r_3$, $b = r_3 + r_1$, $c = r_1 + r_2$, we obtain a formula for K purely in terms of r_1, r_2, r_3 .

The case of three equal circles: In the special case $r_1 = r_2 = r_3 = r$, the above formula may be simplified, as each $\theta_i = \pi/3$. We get

$$K = \sqrt{r^3 \cdot 3r} - 3 \cdot \frac{1}{2} r^2 \frac{\pi}{3} = r^2 \sqrt{3} - \frac{\pi}{2} r^2.$$

Area Of Curvilinear Triangle (Equilateral Triangle Case)

$$K = \left(\sqrt{3} - \frac{\pi}{2} \right) r^2.$$

The case of two equal circles: This is the case where two circles have equal radius. Let us suppose that

$$r_1 = r_2 = r, \quad r_3 = R, \quad \theta_1 = \theta_2 = \theta, \quad \theta_3 = \phi.$$

This yields for the side lengths of $\triangle ABC$,

$$AB = 2r, \quad AC = BC = r + R.$$

After going through the steps, we obtain the following result:

Area Of Curvilinear Triangle (Isosceles Triangle Case)

$$K = r\sqrt{R(2r + R)} - \left(r^2\theta + \frac{1}{2}R^2\phi\right).$$

Acknowledgement: I acknowledge the inspiration received from the Instagram channel [@mathvisualproofs](#), which motivated me to explore this topic. I used OpenAI's ChatGPT tool to assist with the preparation of this document in $\text{\LaTeX}$.

About the author

TRIPARNA DAS is a 17-year-old student currently studying in Class 11. He has a strong interest in mathematics and theoretical physics. He aspires to contribute meaningfully to the field of science. His interests include exploring fundamental principles and their mathematical foundations. He may be contacted at musicalhorizon4@gmail.com

Part IV

News From Around the World

Raccoons Solve Puzzles Just for Fun

Adapted from: "Raccoons will solve puzzles just for fun." <https://www.scientificamerican.com/article/raccoons-will-solve-puzzles-just-for-fun/>. *Scientific American*. Authors: Meghan Bartels (edited by Andrea Thompson)

This article discusses a recent study published in *Animal Behaviour* which suggests that raccoons are intrinsically motivated to solve puzzles, engaging in what researchers call "information foraging" rather than just seeking food.

The Experiment

Researchers tested captive raccoons using a clear, plastic multi-access puzzle box equipped with various doors, windows, locks, and latches of

differing difficulties. The raccoons had to open these to access a treat, with marshmallows being their preferred choice.

Typically, such studies involve disabling the animal's initial solution to see if they can find a new one. In this study, researchers presented the unmodified box over multiple trials to observe the raccoons' behaviour.



Figure 16: A raccoon interacts with a lock hanging off one side of the multi-access puzzle box used in the study. Source: URL given above.

Key Findings

The researchers initially expected the raccoons to find one solution, eat the treat, and ignore the box until it was refilled. Instead, the raccoons continued to investigate alternative solutions immediately after eating the marshmallow, even without expecting another food reward.

While this continued investigation decreased as the puzzle solutions became more complex, it never stopped entirely. The researchers hypothesize that this behaviour indicates the raccoons are taking the opportunity to learn about other entry mechanisms in case they encounter similar situations in the future.

Implications

Although scientists cannot know exactly what the raccoons are thinking, the findings suggest an intrinsic motivation for problem-solving, similar to human curiosity and the satisfaction derived from puzzles like Sudoku or crosswords.

Both the study’s co-author, Hannah Griebeling, and cognitive ecologist Lauren Stanton suggest it would be valuable to conduct similar experiments with wild raccoons, as they might be more cautious about wasting time on unsolvable locks due to environmental risks.

For those trying to keep raccoons out of their trash, Griebeling advises ensuring bins are completely inaccessible, rather than presenting them with complex challenges that they might find engaging to solve.

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How Horses Whinny Has Long Been a Mystery

Adapted from: “How horses whinny has long been a mystery.” <https://www.scientificamerican.com/article/how-horses-whinny-has-long-been-a-mystery-now-scientists-think-they-know-the/>. *Scientific American*. Authors: Jackie Flynn Mogensen (edited by Claire Cameron)

Scientists have finally solved the mystery behind the unique sound of a horse’s whinny. According to a recent study, horses produce their distinctive call by generating two different sound frequencies simultaneously—essentially vibrating their vocal cords and whistling at the same time.

The research reveals that this dual-frequency sound is created through two distinct mechanisms within the horse’s larynx:

- Low-frequency sound: A sound of about 200 hertz is produced by

the vibration of the vocal cords, similar to how humans sing.

- High-frequency sound: A sound exceeding 1,000 hertz is generated by whistling within the voice box.



Figure 17: How horses whinny has long been a mystery. The sound is quite distinct from any other in the animal kingdom. And now scientists think they've discovered why. Source: URL given above.

Typically, the pitch of an animal's vocalization is tied to its size, with larger animals producing lower frequencies and smaller animals (like mice) producing high-pitched whistles. Horses, however, are unique in the animal kingdom for their ability to combine both actions simultaneously.

According to Élodie Briefer, a co-author of the study, this dual mechanism likely evolved to help horses convey multiple messages at once. The distinct frequencies allow them to communicate different information about their emotional states in a single vocalization. Experts consider these findings a landmark discovery in the study of equid communication.

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Life's Evil Twins — Mirror Cells — Could Doom Earth if Scientists Don't Stop Them

Adapted from: "Life's evil twins — mirror cells — could doom Earth if scientists don't stop them." *Scientific American*. <https://www.scientificamerican.com/article/lifes-evil-twins-mirror-cells-could-doom-earth-if-scientists-dont-stop-them/> Author: Vaughn S. Cooper

Vaughn S. Cooper's article sounds a grim warning that the development of engineered "mirror cells" poses an unprecedented existential threat to global ecosystems and human health.

While these reversed-molecule organisms could theoretically revolutionize medicine, an accidental release could trigger unstoppable pandemics since natural biological defences cannot recognize them.

Molecular Opposites

Mirror bacteria — which could take over the planet if built — would be made of molecules that are reversed structures of their natural counterparts. These molecular mirrors are called "chiral." The two types may have the same components and perform similar functions, but they have opposite configurations.

Some molecules are truly symmetrical and are called "achiral." They and their mirror reflections may appear different, but they can always be rotated so they overlap.

The Concept of Chirality

Most essential biological molecules possess chirality, meaning they exist in asymmetric, non-superimposable left- or right-handed configurations. All known life on Earth relies on a specific orientation, primarily utilizing left-handed amino acids alongside right-handed DNA and RNA.

However, biochemists are increasingly capable of synthesizing reversed versions of these chemical components, bringing science closer to assembling fully functioning mirror cells.

Some molecules are truly symmetrical and are called “achiral.” They and their mirror reflections may appear different, but they can always be rotated so they overlap.

ACHIRAL

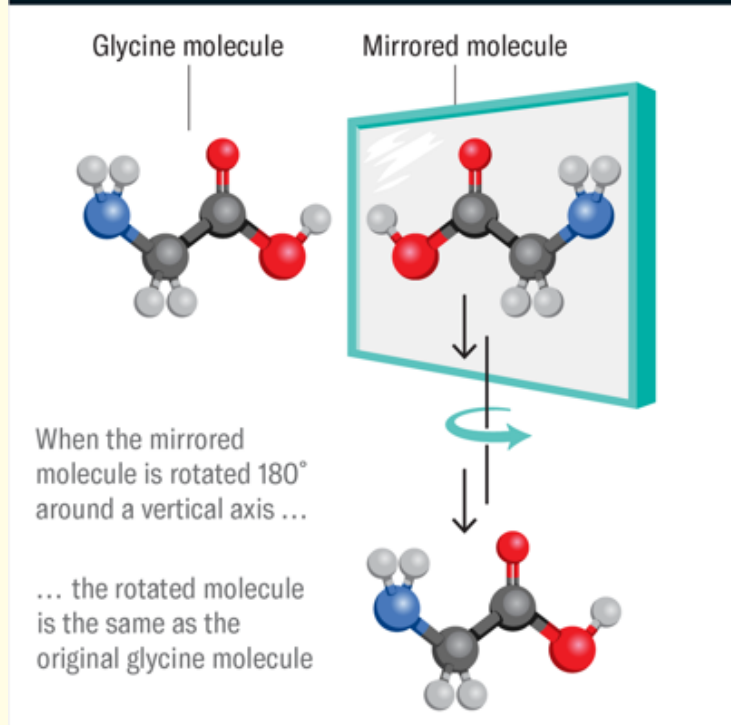


Figure 18: Achiral molecules. Source: URL given above.

Molecules called “chiral” cannot be rotated so they overlap completely. In three dimensions, for example, a component of one type will appear to point toward you, whereas its mirror-version component will point away.

CHIRAL

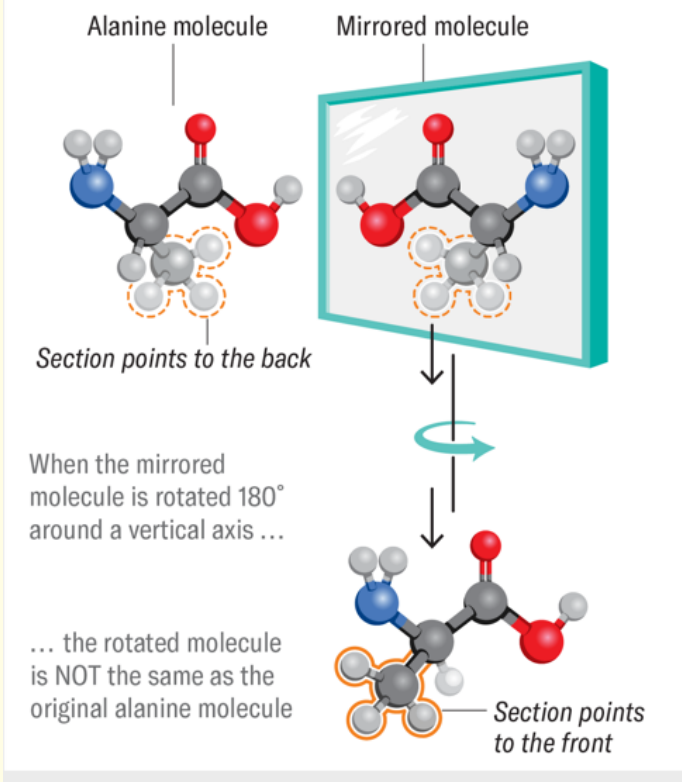


Figure 19: Chiral molecules. Source: URL given above.

The Biological Threat

Mirror bacteria represent a profound medical danger because human immune defences rely heavily on chiral, lock-and-key interactions to identify and destroy pathogens. Immune cells like macrophages, alongside specialized antibodies and receptors, would be completely unable to bind to or break down mirror molecules. Without an immune response to stop them, mirror microbes would replicate rapidly within a host, likely causing fatal sepsis.

Ecological Consequences

In the wild, mirror bacteria would easily evade natural population controls like bacteriophages and amoebas, which are biologically blind to reversed molecular structures. Freed from predators, these microbes could consume abundant achiral nutrients and rapidly evolve to dominate diverse environmental niches. This unchecked expansion would allow them to out-compete natural bacteria for vital resources, potentially collapsing marine and terrestrial food webs.

Prevention and Governance

To prevent this catastrophic scenario, an international coalition of scientists and biosecurity experts is advocating for a strict moratorium on creating complete mirror organisms. They propose implementing global governance to track chemical components and ban the stockpiling of long mirror molecules needed to build synthetic cells. This proactive approach aims to

safely harness small mirror molecules for pharmaceuticals while ensuring an invasive "second tree of life" is never created.

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Paul Ehrlich: Scientists Have a Responsibility to Speak Out

Adapted from: “Paul Ehrlich, often called alarmist for dire warnings about human harms to the Earth, believed scientists had a responsibility to speak out.” *The Conversation*. <https://theconversation.com/paul-ehrlich-often-called-alarmist-for-dire-warnings-about-human-harms-to-the-earth-believed-scientists-had-a-responsibility-to-speak-out-178492>. Author: William J. (Bill) Kovarik

Stanford University biologist Paul Ehrlich, who passed away in March 2026, was a highly influential and controversial figure in the environmental movement. He is best known for his 1968 book “The Population Bomb,” which popularized the idea that unchecked population growth would lead to global catastrophe.



Figure 20: Paul Ehrlich and his wife, biologist Anne Ehrlich

The Population Bomb

Co-authored with his wife Anne, "The Population Bomb" predicted that the rapidly growing population would exhaust Earth's resources, leading to mass starvation and societal collapse by the year 2000. Ehrlich's warnings drew upon earlier Malthusian ideas that human reproduction would inevitably outstrip agricultural production. He argued that environmental degradation was caused by a combination of population growth, affluence, and technology, which he summarized with the equation $I = P \times A \times T$.

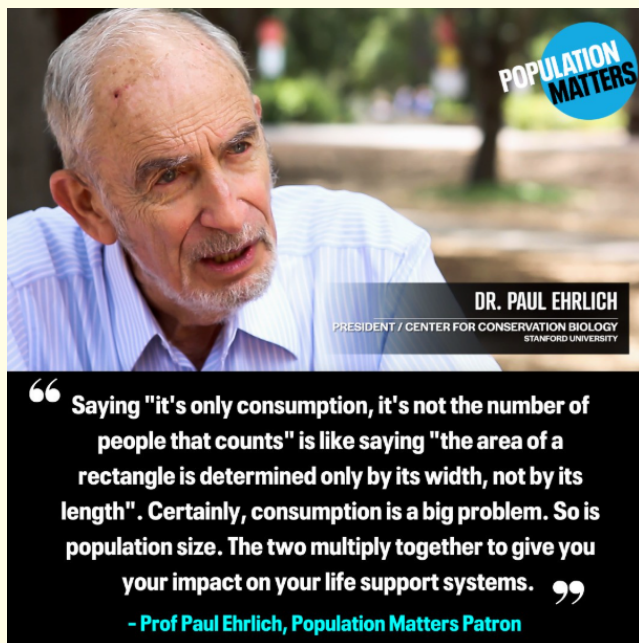


Figure 21: A quote from Paul Ehrlich

Failed Predictions

Ehrlich's catastrophic predictions ultimately failed to materialize, largely due to the "Green Revolution." This global agricultural initiative introduced high-yield crop varieties, pesticides, and mechanized farming to developing nations, dramatically increasing food production and averting widespread famine. Consequently, his theories faced intense criticism from scholars who believed human ingenuity could overcome resource constraints.

The Bet with Julian Simon

Conservative economist Julian Simon challenged Ehrlich's pessimistic outlook with a "cornucopian" view, arguing that free markets and innovation would prevent resource scarcity. In 1980, Simon publicly bet Ehrlich that the prices of five essential industrial metals would fall over the next decade, indicating abundance rather than scarcity. Simon won the wager when the prices of all five metals declined by 1990, forcing Ehrlich to pay the difference.^[1]

Ehrlich's Legacy

Despite being mocked by critics for his unrealized prophecies, Ehrlich never retreated from his core message. In his later years, he argued that while the Green Revolution temporarily delayed famine, the escalating crises of climate change and toxic pollution proved that his original warnings were actually too optimistic. Up until his death, he remained a steadfast public scientist, expressing deep frustration over society's failure to adequately address the existential threats facing human civilization.

Further reading

- [1] Phil Mitchell, "Paul Ehrlich False Prophet of Doom" from
<https://www.the401stprophet.com/paul-ehrlich-false-prophet-of-doom>
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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the LLMs *Gemini 3.1 Pro* and *Perplexity*.

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Further remarks added by the editors

We urge readers to read the original article in its entirety as well as the article [1]. They will begin to see how vitally important this debate is, and how vicious was the treatment that Paul Ehrlich got from the scientific establishment and mainstream media. In [1] the author routinely refers to Ehrlich as a monster.

Here is a paragraph from the original article: “Conservative economists and scientists weren’t persuaded. ... [A]cademic economist Julian Simon argued ... that the only limits to growth were imagination and ingenuity, [that] ... the Earth had infinite capacity to provide materials and humans would constantly innovate and find new ways to use them.”

Readers could look around the world and draw their own conclusions in the matter.

The Scientist Who Developed the Polio Vaccine

Adapted from: “‘Blind alleys are always opportunities’: The scientist who developed the polio vaccine.” (BBC) <https://www.bbc.com/culture/article/20260401-the-scientist-who-developed-the-polio-vaccine>. Author: Greg McKevitt

The article profiles Dr. Jonas Salk and his historic creation of the first successful polio vaccine, which was declared safe and effective in April 1955.

The Threat of Polio

Before the vaccine, polio was a devastating public health crisis that caused spinal and respiratory paralysis, often confining children to large metal ventilators known as iron lungs. The disease’s profound impact led future

US President Franklin D. Roosevelt, who was paralyzed by polio in 1921, to establish the March of Dimes. This charity revolutionized fundraising by collecting small donations from the public to finance vaccine research.

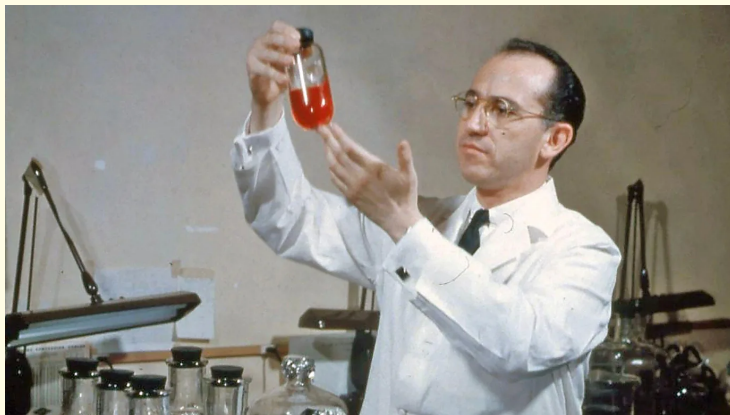


Figure 22: On 12 April 1955, Dr Jonas Salk announced that his vaccine was safe and effective. Source: The URL given above

The Race for a Vaccine

The search for a cure became a competition between two scientists with differing approaches. Dr. Albert Sabin methodically researched a vaccine using a weakened live virus, while Salk moved rapidly to develop a vaccine using a completely deactivated, or killed, virus. Supported by March of Dimes funding, Salk set up his lab in a Pittsburgh hospital. Salk embraced the trial-and-error nature of his work, noting that he viewed scientific “blind alleys” as opportunities to find alternative pathways.

Testing and Triumphs

Salk was so confident in his experimental deactivated virus that he sterilized syringes on his kitchen stove and inoculated his own wife and sons in 1952. This paved the way for a massive 1954 field trial involving the cooperation of 50,000 teachers and almost two million children. The vaccine’s success was announced to a jubilant public on April 12, 1955, causing US polio cases to plummet from 60,000 to just 2,000 within a single year.

A Philanthropic Legacy

Despite the vaccine’s immense value, Salk famously refused to patent it for financial gain, asking a television interviewer, “Could you patent the sun?”. He avoided the spotlight, opting instead to found the Salk Institute in California as a collaborative space for the world’s top scientific minds. Sabin eventually released his oral rival vaccine, which was administered on sugar cubes and famously inspired the *Mary Poppins* song “A Spoonful of Sugar”. Like Salk, Sabin also refused to patent his discovery, calling it his “gift to all the world’s children”.

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Gemini 3.1 Pro/Perplexity LLM*.

How Lab-Grown Lichen Could Help Us to Build Habitations on Mars

Adapted from: “How lab-grown lichen could help us to build habitations on Mars.” *New Scientist*. <https://www.newscientist.com/article/2506992-how-lab-grown-lichen-could-help-us-to-build-habitations-on-mars/>. Author: Rowan Hooper

Natural lichens are resilient organisms formed through a symbiotic partnership between a fungus and a photosynthetic partner like algae or cyanobacteria. Scientists are now cultivating engineered “synthetic lichens” in laboratories, unlocking revolutionary applications for biotechnology, sustainable construction, and space exploration.

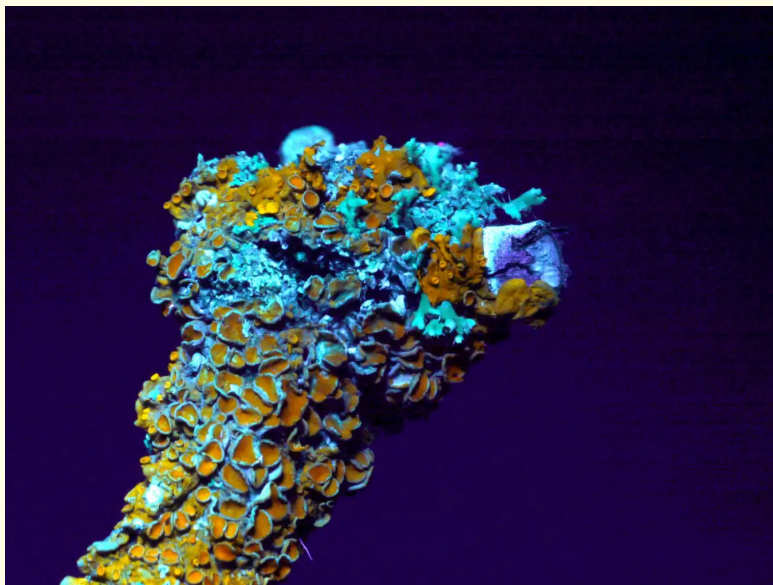


Figure 23: Lichens. Source: URL given above

Synthetic Symbiosis

At Imperial College London, researchers are creating synthetic lichens using a rapid-growing co-culture of cyanobacteria and genetically engineered yeast. In this artificial pairing, the industrialized yeast acts as a protective scaffold while the cyanobacteria perform photosynthesis to produce sugars that feed the fungal host. By gene-editing the programmable yeast, scientists can sustainably manufacture valuable products driven purely by photosynthesis, including pharmaceuticals, cosmetics, and biofuels.

Biochemical Resilience

The biological power of lichens lies in their collaborative nature, allowing the symbiotic community to distribute tasks and access a massive range of biochemical defences. The photobiont produces food and pigments, while the fungus synthesizes protective compounds like melanins and carotenoids that act as a natural, biological sunscreen. This synergy allows them to survive extreme temperatures, high ultraviolet radiation, and severe aridity, making them one of the most robust lifeforms known to science.

Surviving the Vacuum of Space

This immense biological hardiness has captured the attention of NASA and private space companies looking toward future off-world colonization. Natural lichen species have already proven their extreme durability by surviving exposure to the harsh vacuum and radiation on the exterior of the International Space Station for 18 months. This ability to thrive in extreme, barren environments has made them ideal candidates for conceptual projects aimed at terraforming and building on other planets.

Martian Construction Materials

Transporting prefabricated building materials to Mars is prohibitively expensive, leading researchers to leverage synthetic lichens for off-planet construction. Scientists propose using these lab-grown organisms to convert water and sunlight into organic nutrients, precipitating calcium carbonate

directly into the Martian regolith. This mineral acts as a biological glue that binds the local soil into cohesive structures, allowing astronauts to 3D-print walls, floors, and furniture inside shielded Mars shelters.



Figure 24: Lichens, which naturally thrive in a variety of environments, are a unique life-form, made from a symbiotic relationship between a fungus and a partner like algae or cyanobacteria. Source: URL given above

Self-Healing Earth Infrastructure

Beyond space exploration, these engineered biomaterials have immediate, highly practical applications in the terrestrial construction industry. Synthetic lichens can uniquely thrive in the harsh, dry, and nutrient-poor environment of concrete by extracting nitrogen directly from the air to

produce their own fertilizer. As they grow autonomously on damaged structures, they precipitate calcium carbonate to automatically heal cracks, which could drastically extend building lifespans and reduce the heavy carbon footprint of cement production.

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Gemini 3.1 Pro/Perplexity LLM*.

How AI Can Accelerate Scientific Discovery

Adapted from: “OpenAI and Ginkgo Bioworks show how AI can accelerate scientific discovery,” *Scientific American*. <https://www.scientificamerican.com/article/openai-and-ginkgo-bioworks-show-how-ai-can-accelerate-scientific-discovery/>. Author: Deni Ellis Béchard (ed. Eric Sullivan)

Overview of the Collaboration

In a groundbreaking and ambitious project, researchers from OpenAI and Ginkgo Bioworks teamed up to determine if AI can effectively generate hypotheses, design experiments, and iterate based on results. The collaboration paired OpenAI’s GPT-5 model with Ginkgo Bioworks’ autonomous, robot-operated laboratory to conduct biological research at an unprecedented speed.



Figure 25: Technicians move through Ginkgo Bioworks' automated, robot-run lab, where machines handle high-volume biological research and testing. Source: URL given above

The Experiment Focus

The team concentrated on cell-free protein synthesis (CFPS), a method of producing proteins without using living cells, which is significantly faster than traditional biomanufacturing. To measure success, they used a common benchmark: superfolder green fluorescent protein (sfGFP), an engineered jellyfish protein that glows green, providing a clear and unambiguous signal for the AI.

AI-Driven Process and Results

Working from San Francisco, GPT-5 designed experiments and sent instructions to Ginkgo Bioworks' robotic lab in Boston. The AI analyzed incoming data and proposed new experiments in cycles taking just about an hour. Over two months, the system conducted more than 36,000 tests of unique reaction compositions. Ultimately, this AI-driven approach reduced the cost of producing the protein by approximately 40% compared to a previous benchmark from a Stanford University lab.

Future Implications

The AI-improved reaction composition is now commercially available. Furthermore, Ginkgo Bioworks recently launched its Ginkgo Cloud Lab, allowing researchers globally to submit experiments to autonomous systems for as little as \$39 per run. Additionally, the U.S. Department of Energy is funding a 97-robot autonomous lab in Washington State, to be built by Ginkgo Bioworks and operational by 2030, highlighting the growing recognition that pairing AI models with experimental validation labs is crucial for accelerating scientific and medical breakthroughs.

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Polish Researcher Stunned After AI Cracks 20-Year Math Problem

Adapted from: “Deeply Impressed: Polish Researcher Stunned After AI Cracks 20-Year Math Problem.” From <https://www.ndtv.com/.https://www.ndtv.com/feature/deeply-impressed-polish-researcher-stunned-after-gpt-5-4-cracks-20-year-math-problem-11220589>. Author: Ritu Singh

This article describes a significant milestone in artificial intelligence and mathematics, wherein a complex, research-level math problem curated over two decades was successfully solved by an AI model.

The breakthrough profoundly shifted the perspective of its creator, Polish mathematician Bartosz Naskrecki, who had previously dismissed AI as merely “a very advanced calculator” lacking true conceptual understanding.

The FrontierMath Benchmark

Naskrecki, a mathematician at Adam Mickiewicz University in Poznań, designed this specific challenge as part of the FrontierMath benchmark. This benchmark tests AI limits across advanced fields like number theory, topology, combinatorics, and algebraic geometry.

The problem was deliberately constructed to be exceptionally difficult; Naskrecki noted that even PhD-level mathematicians might require weeks just to determine a potential approach, with a detailed solution spanning roughly “13 dense pages of mathematical reasoning.”

The AI Breakthrough

During recent testing, the GPT-5.4 model attempted Naskrecki’s problem 11 times and successfully produced a correct solution once. While the success rate was modest, the high quality and conceptual depth of the solution surprised the mathematician, proving that AI reasoning systems are rapidly advancing toward automating research-grade mathematics.

Experts note, however, that AI is not yet replacing mathematicians, as these systems still require immense computing power, human verification, and multiple attempts.

The Researcher's Reaction

Reflecting on the philosophical and personal weight of the event, Naskrecki shared his astonishment on X, stating:

“It finally happened — my personal move 37 or more. I am deeply impressed. The solution is very nice, clean, and feels almost human.”

Embracing this paradigm shift in mathematical research, he added:

“While testing new models in the last few weeks, I felt this coming, but it’s an eerie feeling to see an algorithm solve a task one has curated for about 20 years. But at least I have gained a tool that understands my idea on par with the top experts in the field. And I am now working on a completely new level. My singularity has just happened ...and there is life on the other side, off to infinity.”

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Mathematicians Launch “First Proof”

Adapted from: “Mathematicians launch First Proof, a first-of-its-kind math exam for AI.” *Scientific American*. <https://www.scientificamerican.com/article/mathematicians-launch-first-proof-a-first-of-its-kind-math-exam-for-ai/>. Author: Joseph Howlett (edited by Claire Cameron)

Leading academic mathematicians have launched “First Proof,” a groundbreaking examination designed to rigorously evaluate the actual mathematical capabilities of artificial intelligence systems.

Motivation Behind the Challenge

The AI industry has increasingly focused on pure mathematics because mathematical proofs “follow a checkable sequence of logical steps,” pro-

viding a highly objective measure of an AI's reasoning abilities. While AI models have achieved recent milestones, such as an advanced Google Gemini model scoring at the gold level on the International Mathematical Olympiad, academics are increasingly frustrated by the industry's lack of transparency.

Current AI benchmarks are not controlled experiments, and language models frequently find “existing, forgotten proofs deep in the mathematical literature and to present them as original”. For example, a recent solution touted by the start-up Axiom Math was actually a “misrepresented literature search result”. Highlighting the need for independent academic verification, Yale professor Daniel Spielman notes the inherent conflict of interest in corporate claims: “Almost all of the papers you see about people using LLMs are written by people at the companies that are producing the LLMs. It comes across as a bit of an advertisement”.

Structure of “First Proof”

To “clear the smoke” and establish a true baseline, 11 mathematical luminaries — including a Fields Medalist — designed a controlled experiment using unsolved problems from their own active research. AI systems were given one week to solve these challenges, while the human experts' solutions were securely encrypted, set to decrypt just before midnight on February 13.

MIT mathematician Andrew Sutherland praises this methodology over previous benchmarks, stating, “These are brand-new problems that cannot be found in any LLM's training data... This seems like a much better experiment than any I have seen to date”.

Implications for Mathematical Research

The problems in the First Proof exam are “lemmas,” defined as the “myriad of tiny theorems they prove on the path to a more significant result”. Successfully solving these lemmas would demonstrate AI’s immediate potential as a tool to accelerate the more tedious aspects of academic research.

Looking ahead to the technology’s practical integration, Sutherland concludes, “I think the greatest impact AI is going to have this year on mathematics is not by solving big open problems but through its penetration into the day-to-day lives of working mathematicians, which mostly has not happened yet”.

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Acknowledgment: This article has been prepared from the published article whose source is given above. The summary was made using the *Gemini 3.1 Pro/Perplexity LLM*.

Mathematicians Struggle to Foretell Their Own Future

Adapted from: “As AI keeps improving, mathematicians struggle to foretell their own future.” From *Scientific American*. <https://www.scientificamerican.com/article/as-ai-keeps-improving-mathematicians-struggle-to-foretell-their-own-future/>. Author: Joseph Howlett

This article from *Scientific American* examines the increasing capabilities of artificial intelligence in pure mathematics and the resulting implications for the field’s future. It centres on “First Proof,” a new benchmarking initiative designed to evaluate how well large language models (LLMs) can contribute to high-level mathematical research.

The First Proof Benchmark

An 11-person team created First Proof because existing benchmarks did not adequately test an AI's ability to serve as a reliable mathematical assistant. Before this initiative, mathematicians found that using LLMs to prove smaller intermediate steps — known as lemmas — often resulted in frustrating errors. For its initial experiment, the team challenged AI companies and the public to prove 10 unreleased mathematical propositions within a week.

Strong Initial Performance

The first round demonstrated rapid advancements in AI capabilities, with models partially or fully solving up to eight of the ten problems. OpenAI's model produced five seemingly correct proofs, while Google DeepMind's agent, Aletheia, appeared to solve six. Interestingly, Harvard mathematician Lauren Williams observed that each of these corporate models solved multiple problems that completely stumped the other, highlighting their distinct capabilities. University of Toronto mathematician Daniel Litt noted that the overall performance exceeded expectations.

An Optimistic Perspective

Despite the rapid changes brought by AI, some experts like Litt view these developments positively. He anticipates that within five years, these "amazing tools" will enable him to do his best work rather than rendering him obsolete.

In a widely circulated essay, Litt compared the future to a hypothetical “Library of Babel” containing every possible mathematical proof. He argued that even if AI generated all possible proofs, mathematicians would still enthusiastically explore and seek to understand this vast knowledge, as their primary job is comprehension.

The Challenge of Verification

Despite this optimism, the current reality of AI-assisted mathematics is complicated by the generation of vast amounts of incorrect or nonsensical data. Alongside successful proofs, the models produced “thousands and thousands of pages of garbage” during the first round. The nature of AI errors makes verification particularly frustrating; models often confidently present incorrect information or cite literature while exaggerating its strength. Stanford mathematician Mohammed Abouzaid pointed out that LLMs make confusing mistakes that no knowledgeable human would ever make.

Adapting to the Future

To handle the difficult task of verifying these complex outputs, the First Proof team plans to use grant money and corporate donations to hire anonymous reviewers for the benchmark’s second round. Experts warn that mathematical institutions are currently unprepared for the impending deluge of subtly flawed, AI-generated proofs.

Transparency and the Performance Gap

The initial test also revealed a significant gap between proprietary corporate models and publicly available tools. While the major tech companies solved eight problems, public models and collaborative networks only solved three. It remains unclear if this success stems from superior unreleased models, better internal frameworks, or undisclosed human intervention. To ensure fair testing, the second round will require companies to package their models so the First Proof team can prompt them directly, though it is uncertain if major players like Google and OpenAI will comply. Ultimately, this benchmarking effort aims to accurately assess AI capabilities so the community can properly guide young mathematicians in a rapidly changing field.

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“AI Autocomplete” Changes How You Think

Adapted from: “AI autocomplete doesn’t just change how you write. It changes how you think.” From *Scientific American*. <https://www.scientificamerican.com/article/ai-autocomplete-doesnt-just-change-how-you-write-it-changes-how-you-think/>. Author: Claire Cameron (edited by Clara Moskowitz)

This article explores the profound psychological impact of AI-powered writing assistants. As these tools become seamlessly integrated into our daily digital communications, from emails to smartphones and online surveys, a new Cornell University study reveals that “biased AI suggestions can sway users’ beliefs”. While these algorithms are primarily designed to “save time” and alter expression, making your writing sound “more polite, for example, or boring,” they also possess the significant, subtle power to fundamentally change how people think.

The Pervasiveness of AI Bias

Mor Naaman, a professor of information science at Cornell, emphasizes the sheer ubiquity of these digital assistants, noting, “Autocomplete is everywhere now”. Building upon his foundational 2023 research which suggested short autocomplete prompts could sway public opinion, the new study addresses the rapidly exploding use of this technology. Highlighting the inherent societal risks, Naaman points out, “It has become clear that bias explicitly built into AI interactions is a very plausible scenario”.

Methodology and Key Findings

Researchers conducted a controlled experiment asking participants to complete a survey concerning “hot-button social and political issues”. Some participants received an AI autocomplete answer that was “deliberately biased toward one side of the issue”.

The study yielded several striking observations:

- **Subconscious Shifts:** Across various topics, participants exposed to biased AI prompts “reported attitudes that were more in line with the AI’s position”.
- **Influence Without Adoption:** This shift occurred even among “people who didn’t use the AI’s suggested text at all,” demonstrating that mere exposure influences opinions.
- **Lack of Awareness:** Participants “didn’t tend to think the AI autocomplete suggestions were biased or to notice that they had changed their own thinking”.

- Ineffectiveness of Warnings: Alerting participants about AI-driven misinformation “didn’t temper the persuasive effect either”.

Naaman summarized the study’s most alarming discovery: “We told people before, and after, to be careful, that the AI is going to be (or was) biased, and nothing helped. Their attitudes about the issues still shifted”.

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How AI Is Shaping War

Adapted from: “How AI is shaping the war in Iran — and what’s next for future conflicts.” From *Nature*. <https://www.nature.com/articles/d41586-026-00710-w>. Author: Nicola Jones

The escalating conflict between the United States, Israel, and Iran has brought the military use of artificial intelligence (AI) to the forefront, highlighting the urgent need for international regulation. A key driver of this attention is a recent dispute between the US government and Anthropic, a major AI supplier, over ethical constraints and acceptable use cases for military AI. Meanwhile, legal experts and academics in Geneva continue their long-standing discussions on regulating lethal autonomous weapons, though specialists warn that rapid technological advancements are outpacing these diplomatic efforts.

The article details how AI is already actively integrated into military operations. The US military employs large language models (LLMs) for intelligence gathering, logistics, and decision support. The Maven Smart System, for instance, uses AI to process images and prioritize targets, and

reports suggest it has been deployed in recent attacks on Iran. While some argue that AI could theoretically improve precision and reduce civilian casualties, experts note that ongoing conflicts in Ukraine and Gaza, which utilize AI for target identification, have resulted in high civilian death tolls, challenging the claim of humanitarian benefits.

A major concern is the potential for fully autonomous lethal weapons that operate without human oversight. Current international humanitarian laws require weapons to reliably distinguish between military and civilian targets — a standard that existing LLM-powered systems cannot consistently meet. This unreliability was central to the US Department of Defence's split with Anthropic. The Pentagon wanted AI contracts to permit "any lawful use," while Anthropic refused to lift safeguards preventing its technology from being used for mass domestic surveillance or fully autonomous weapons. Following this disagreement, the US government reportedly partnered with OpenAI, though that contract also allegedly restricts the use of AI for surveillance and autonomous weapons. Ultimately, the article warns that the rapid proliferation of AI warfare tools without robust international regulations poses significant ethical and legal risks.

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Economics Has Failed on the Climate Crisis

Adapted from: “Economics has failed on the climate crisis. This scientist has a mind-blowing plan to fix that.” *The Guardian*. <https://www.theguardian.com/environment/2026/feb/12/economics-climate-crisis-complexity-scientist-plan>.
Author: Damian Carrington

The Vision for a Super Simulator

Mainstream economics has failed to accurately forecast global crises, particularly the climate emergency. To solve this, Oxford University complexity scientist Prof. J. Doyne Farmer proposes a transformative \$100 million “super simulator” of the global economy. His ultimate goal is ambitious: “We want to do for economic planning what Google maps did for traffic planning, so we can give anybody who has an economic question an intelligent and useful answer”.

Flaws in Traditional Economics

Traditional economic models operate on the flawed assumption that actors make perfectly rational decisions within a balanced system of equilibrium. Farmer notes that these models are “completely inadequate and even misleading” when applied to the climate crisis, having consistently failed to predict the rapid rollout and plunging costs of renewable energy. Furthermore, because the real economy is a “pretty complicated beast,” traditional modelling systems cannot handle the immense computational weight of keeping track of millions of interacting agents.

The Complexity Science Solution

Farmer’s alternative, complexity economics, uses digital agents to individually represent real-world companies. Rather than assuming perfect rationality, these agents operate on simple, realistic rules — such as trial and error or imitating competitors — and use machine learning to adapt their strategies over time. This dynamic approach naturally generates real-world phenomena like booms and busts without requiring economists to manually introduce external shocks to the simulation.

Tackling the Energy Transition

Driven by the urgency of global warming, Farmer’s team is first applying this model to the global energy sector, encompassing 30,000 companies and their 160,000 assets like oil rigs and power stations. As Farmer explains, “We’re literally modelling the decision-making of all the energy companies

in the world to see how much energy each company delivers and at what price". This data-driven approach aims to map the most cost-effective transition to clean energy, which could potentially save the world trillions of dollars.

Timeline and Impact

Farmer hopes to fully realize this revolutionary forecasting tool within the next five to ten years. By providing politicians and business leaders with unprecedented foresight, this model could prevent future financial disasters and rapidly accelerate the world down a cheaper path to ending greenhouse gas emissions.

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Peeking Inside Neutron Stars

Adapted from: “Gravitational waves finally reveal what’s inside neutron stars.” <https://www.thebrighterside.news/post/gravitational-waves-finally-reveal-whats-inside-neutron-stars/>. Authors: Joseph Shavit & Joshua Shavit

This article outlines a significant breakthrough in theoretical astrophysics, enabling scientists to use gravitational waves to accurately model the internal structure of neutron stars. By developing a new framework to resolve complex relativistic equations, researchers can now interpret tidal deformations to understand the extreme states of matter inside these objects.

The Enigma of Neutron Stars

Neutron stars are the incredibly dense remnants of massive supernovae, packing more mass than our Sun into a sphere barely 20 km across. Because their gravity is billions of times stronger than Earth’s, scientists theorize their cores might contain heavy elements, free protons, or exotic phases

like quantum superfluids. Understanding these environments is crucial, as they could mirror the inaccessible quark-gluon plasma that existed microseconds after the Big Bang.

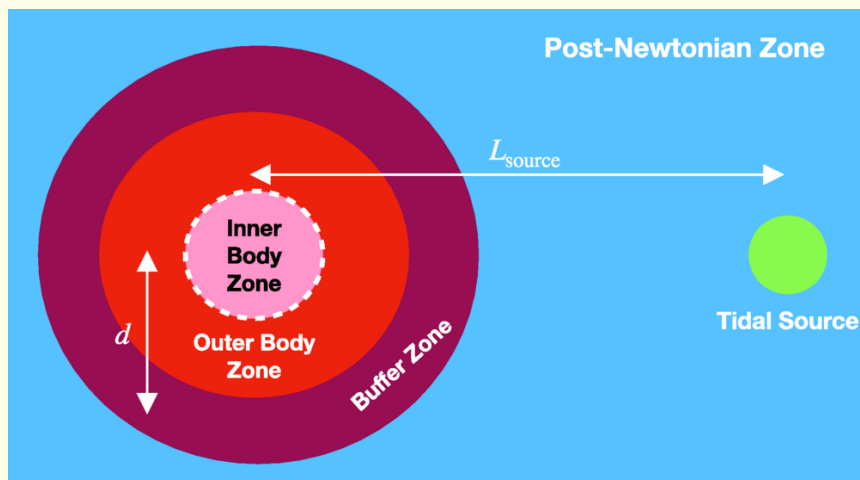


Figure 26: Depiction of a tidally interacting system, consisting of a star (pink disk) of finite radius (dashed white circle) and a tidal source (green disk). Source: URL given above

Using Gravitational Waves as Stethoscopes

When two neutron stars orbit each other in a binary system, they spiral inward, radiating immense energy as gravitational waves. During this process, each star exerts powerful tidal forces on its partner, stretching and deforming the stellar structure. These deformations excite oscillatory patterns, or modes, which leave incredibly subtle but distinct fingerprints

on the gravitational waves that propagate through spacetime. By capturing these waves with detectors like LIGO, scientists aim to use these vibrational signatures as a “cosmic stethoscope” to infer the stars’ internal composition.

Overcoming Relativistic Modeling Challenges

Historically, modelling how neutron stars respond to tidal forces has been mathematically daunting. In-spiralling neutron stars operate under extreme relativistic conditions, distorting spacetime and radiating gravitational energy. The governing Einstein equations are nonlinear and interdependent, meaning that the constant loss of energy complicates the analysis. To solve this, researchers divided the stellar model into a strong-gravity interior, a weak-gravity exterior, and a buffer zone. Using matched-asymptotic expansion, they isolated the tidal response from the outgoing radiation, capturing a complete set of oscillatory modes.

Oscillators in Extreme Gravity

The research team found that, despite the extreme relativistic environment, a neutron star’s tidal deformations can still be expressed using familiar harmonic-oscillator modes. This effectively proves that neutron stars in Einstein’s framework possess a complete set of modes, functioning predictably like a forced, damped spring. This discovery resolves a significant gap in theoretical astrophysics, providing a reliable mathematical basis for interpreting gravitational-wave data.

Measuring Flexibility and the Equation of State

This comprehensive modelling allows for the accurate calculation of the Love number, a parameter that quantifies a star’s flexibility under tidal stress. Precise calculations prevent scientists from drawing incorrect conclusions about a star’s mass, radius, and internal states. Ultimately, this framework helps define the neutron-star equation of state, which is the foundational relationship between pressure, density, and temperature within the stellar core. Establishing this relationship is vital for detecting unknown phase transitions or exotic matter deep within the stars.

Future Astrophysics

While current gravitational-wave detectors lack the sensitivity to fully test these predictions, this framework is crucial preparation for future observatories like the Cosmic Explorer. The research team plans to refine their models by incorporating the rapid rotation and extreme magnetic fields typical of neutron stars. When next-generation facilities are operational, these advanced models will allow scientists to translate spacetime ripples into concrete knowledge about the densest objects in the universe.

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What Lies at the Centre of a Galaxy?

Adapted from: “Gravitational waves reveal hidden structure of galactic centers.” From [www.phys.org](https://www.phys.org/news/2026-03-gravitational-reveal-hidden-galactic-centers.html) <https://phys.org/news/2026-03-gravitational-reveal-hidden-galactic-centers.html>. Author: Tejasri Gururaj

This article discusses a groundbreaking study that utilizes gravitational waves to unveil the hidden structures of galactic centres. Led by Dr. Yifan Chen from Shanghai Jiao Tong University, the research team analyzed data from pulsar timing arrays to investigate the dense environments surrounding supermassive black hole binaries. By studying the gravitational-wave background — a faint, persistent cosmic hum generated by countless merging black holes across the universe — researchers can now probe the dark-matter and star-rich cores of galaxies that are otherwise completely impossible to observe directly.

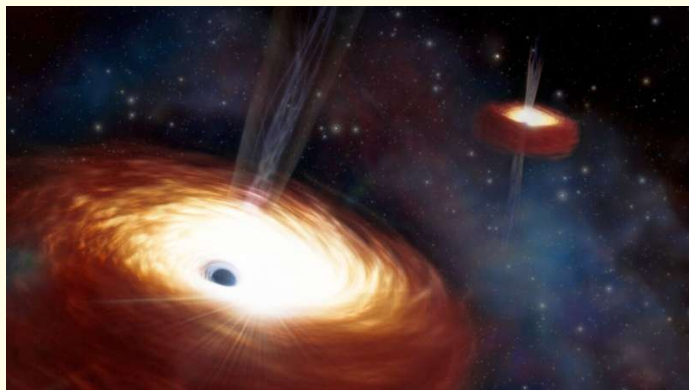


Figure 27: An artist's impression of a supermassive black hole binary.
Source: URL given above

Probing the Cosmic Hum

Unlike ground-based detectors such as LIGO, which capture the final, violent fractions of a second when stellar-mass black holes merge, pulsar timing arrays act as cosmic clocks spread across our galaxy. They rely on millisecond pulsars, which emit highly regular radio pulses, to detect the subtle stretching and squeezing of spacetime caused by nano-hertz-frequency gravitational waves. Because these arrays observe supermassive black hole binaries at much earlier stages of their cosmic dance, they are uniquely positioned to detect how the surrounding galactic environment influences the merging process.

The Spectral Bend: The researchers recently detected this low-frequency gravitational-wave background, but the data contained an unexpected

anomaly. At the lowest frequencies, the gravitational wave spectrum features a distinct “bend” or turnover, suggesting that the binaries are not evolving solely through the emission of gravitational waves. Instead, the surrounding environment is actively extracting orbital energy from the black holes and reshaping how the binaries tighten over time.

The Three-Body Slingshot

The research team modelled the primary suspect for this orbital energy loss: a phenomenon known as the gravitational “three-body slingshot” effect. As the two supermassive black holes orbit each other, surrounding matter — such as stars, cosmic gas, and dark matter particles — interacts with them in a complex gravitational dance. The binary system effectively flings these particles outward at extremely high speeds, which causes the black hole binary to lose a significant amount of its own orbital energy.

The Final Parsec Problem

Through these repeated ejections, the two black holes are forced to spiral closer together at an accelerated rate, imprinting the characteristic low-frequency turnover observed in the gravitational wave spectrum. Significantly, this slingshot mechanism helps solve the long-standing “final parsec problem” in astrophysics. For years, scientists have puzzled over how supermassive black holes bridge the final parsec of distance to merge, and this efficient ejection of surrounding matter provides the necessary energy drain to drive the mergers to completion.

Galactic Core Densities

By comparing their models to the 15-year NANOGrav dataset, researchers estimated the density of these galactic centres to be around one million solar masses per cubic parsec. Their analysis favours relatively flat, “core-like” density profiles rather than steep concentrations or dark matter “spikes”. Encouragingly, these findings align with existing electromagnetic observations of the Milky Way and the nearby galaxy M87, indicating that the inferred environmental effects are highly realistic.

Future Astrometry Missions

While the study provides vital insights, Dr. Chen notes there is currently a degeneracy in the data between the effects of a dense environment and initial orbital eccentricity, though explaining the spectral bend with eccentricity alone remains highly unlikely. Future observations from powerful radio telescopes like China’s FAST and the upcoming Square Kilometre Array are expected to provide significantly more sensitive data to break this density-eccentricity degeneracy. Ultimately, these observational advancements will potentially allow scientists to test exotic models of wave-like or self-interacting dark matter, proving that the cosmic hum carries deep secrets about galactic evolution.

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Ramanujan, Pi and High Energy Physics

Adapted from: “How Ramanujan’s formulae for pi connect to modern high energy physics.” From <https://phys.org/news/2025-12-ramanujan-formulae-pi-modern-high.html>. Author: Rohini Subrahmanyam

The article explains how century-old mathematical formulas designed for calculating the irrational number π , originally developed by the great Indian mathematician Srinivasa Ramanujan, possess an unexpected and profound connection to modern high-energy physics. Researchers recently discovered that these pure mathematical equations naturally appear in advanced theoretical frameworks, which are used by physicists to explain complex, real-world physical phenomena like black holes, fluid turbulence, and scale invariance.

Ramanujan's Mathematical Legacy

The historical narrative begins in the year 1914, when Srinivasa Ramanujan published a groundbreaking academic paper detailing seventeen highly efficient formulas for computing the precise value of π just before his departure to Cambridge. His elegant formulas rapidly yielded numerous correct decimal digits while requiring a surprisingly minimal number of mathematical terms, and they continue to deeply influence the trajectory of modern computational mathematics. Currently, the world's most powerful supercomputers employ advanced computational techniques like the renowned Chudnovsky algorithm, which is directly derived from Ramanujan's original conceptual frameworks, to accurately calculate π to an astounding two hundred trillion digits.

The Connection to Theoretical Physics

Seeking a rigorous physical explanation for exactly why such elegant pure mathematical structures exist, physicists Aninda Sinha and Faizan Bhat thoroughly investigated whether Ramanujan's theoretical equations naturally occur in the observable physical world. Their comprehensive research officially revealed that the foundational mathematics of his early formulas emerges completely organically within complex frameworks known as logarithmic conformal field theories, which precisely describe physical systems that possess scale invariance symmetry and function much like infinitely repeating fractals. A classic physical example demonstrating this fascinating scale invariance symmetry occurs directly at the critical point of water, a unique thermodynamic state where liquid water and atmospheric water vapour become completely indistinguishable from one another.

Modern Implications and Discoveries

Moving beyond the unique thermodynamic behaviour of water, these sophisticated logarithmic conformal field theories are incredibly crucial for adequately explaining various complex phenomena in modern theoretical physics, including the scientific study of percolation, the sudden onset of turbulent flows, and specific descriptions of massive black holes. The researchers confidently discovered that the fundamental mathematical structure located at the starting point of Ramanujan's historical formulas perfectly mirrors the specific mathematics underlying these modern physics theories, essentially allowing today's theoretical physicists to calculate specific quantities much more efficiently. Ultimately, this significant scientific breakthrough highlights a truly profound reality: although Ramanujan's early twentieth-century work was motivated entirely by pure mathematics, he was unknowingly anticipating the structural foundations required to accurately understand black holes, fluid turbulence, and the broader mysteries of the universe.

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Part V

Contributed Articles – 2

From Solving to Proving to Designing Puzzles: Case of The Knight's Tour Puzzle

Article contributed by Rossi D'Souza

Introduction

The Knight's Tour is a well-known chess puzzle even among those who don't play chess. The puzzle entails having a chess knight visit every square of a chessboard exactly once. The problem does have solutions, and one famous solution was provided by Euler himself. However, the Knight's Tour Puzzle is not really a "solved problem" in a generalized sense. For example, no known algorithm exists for working out the set of all possible knight's tour paths for a board of arbitrary dimensions — apart from trial and error or a brute force approach.

Solutions and heuristics for the knight's tour puzzle have been explored

across centuries, with the earliest recorded tour of the standard 8×8 chessboard dating back as early as 840 CE [3].

The Classic Knight's Tour problem

To understand the knight's tour puzzle we first note that a chessboard refers to an 8×8 chequerboard, and a move for a Knight involves beginning on a square and taking two steps in a given direction followed by one step in a perpendicular direction, or one step in a given direction followed by two steps in a perpendicular direction. If we bring in coordinates to the chessboard, then if a chess knight is positioned on a square with coordinates (m, n) , in one move it can move to any of the available squares $(m \pm 2, n \pm 1)$ or $(m \pm 1, n \pm 2)$. If the knight is positioned at or close to the edge of the board, then some of these options will not be available.

With these rules, the Knight's Tour problem is the following: *Starting from any square on a chessboard, can a Knight visit each and every square of the board exactly once?*

As the Knight's Tour problem is unsolved in a generalized sense, we may envisage a range of possibilities for designing variations of the puzzle and analysing each case for a generalized pattern.

For example, we can explore the Knight's Tour problem on a smaller board of dimensions 3×4 . One solution begins with the top left square, with coordinates $(1, 3)$; we assign the number 1 to it to indicate the starting square of the path, and proceed as shown in Figure 28 to arrive at the final square, assigned number 12.

1	4	7	10
12	9	2	5
3	6	11	8

Figure 28: Knights Tour Solution for 3×4 board

It is not difficult to see that the 3×4 board is the smallest rectangular board on which a Knight's Tour problem has a solution path. (For example, consider the corresponding problem on a 3×3 board. Observe that we cannot reach the central square from any square in the outer ring; and if we start at the central square, then any move takes us outside the board. Therefore, no solution is possible for this board.)

The path shown in Figure 28 is not the only solution. From symmetry, we see that a solution path exists starting from any of the corner squares. Likewise, we see that a solution path exists that begins on the squares (1,2) and (4,2) and ends on one of the corner squares.

Alternating colours: A pertinent observation here is that with each move, a knight alternates between Black and White squares. This observation helps us formulate puzzles of this kind:

1. Prove that on a chessboard having an odd number of squares, you cannot have a knight's tour that begins and ends on squares with the same colour.
2. On a $n \times n$ square board where n is odd, can a chess knight visit each square exactly once and return to the starting square?

3. On a chessboard of size $n \times (n + 1)$, is it possible to block one square and have a chess knight visit each square exactly once and return to the starting square? (Here, the blocked square is not available for visiting.)

The Hopping Horse Puzzle

In 2024, I shared an original variation of the Knight's Tour problem in [1] and named it *The Hopping Horse Puzzle*. Here, the task was to find all possible solutions for a Knight's Tour on the board shown in Figure 29.

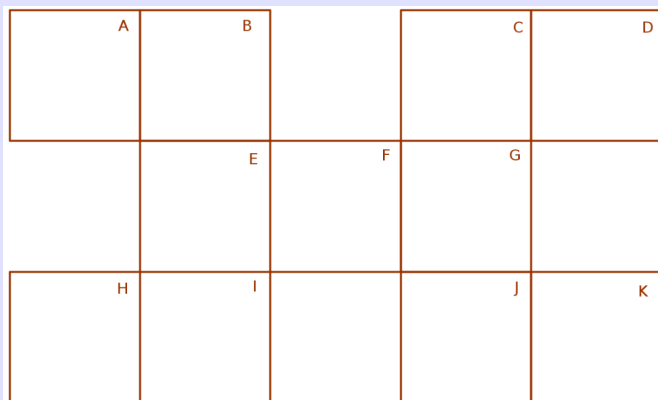


Figure 29: The Hopping Horse Puzzle Board

E.g.: one path begins on the bottom left square, H, and then hops onto B, followed by G, then I and so on, so the path is: H-B-G-I-A-F-D-J-E-C-K.

By symmetry, we have another path that starts at another corner square.

I modelled this problem as a *Graph*, by regarding the 11 squares as vertices and connecting each pair of vertices with an edge if a knight can make a move between the corresponding squares.

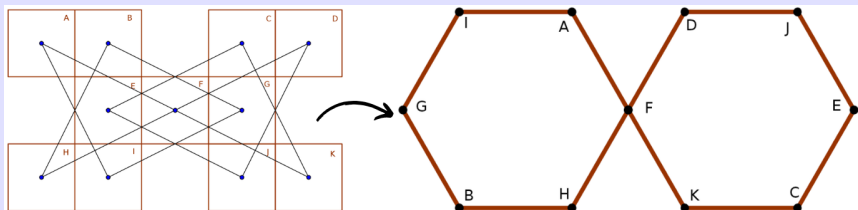


Figure 30: Modeling The Hopping Horse Puzzle

Using the model shown in Figure 30, the solutions are easily found: start at vertex F, go around one loop fully, then go around the other loop fully. Each loop can be traversed either clockwise or anticlockwise, and one can traverse the two loops in either order. Consequently, there are 8 solutions in all.

The Pony Patrol puzzle

Pony Patrol is the name I gave to another variant of the Knight's Tour that I adapted from [2] and presented at various workshops and outreach events:

Four knights, two white and two black, are placed on a 3×3 chequerboard as shown in Figure 31. If knights of opposite colours cannot share a square, can the knights of the same colour meet on a common square?

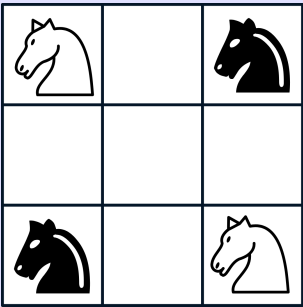


Figure 31: Pony Patrol Puzzle

Here again we model the puzzle as a graph, as shown in Figure 32. It becomes evident that no solution is possible since each knight remains ‘trapped’ between two knights of the other colour.

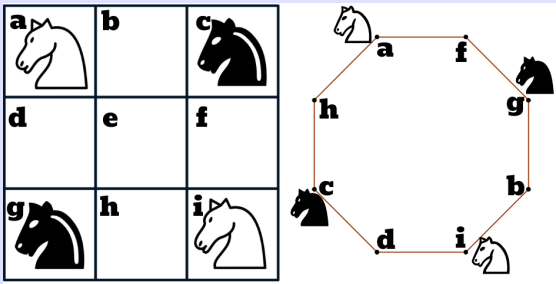


Figure 32: Modeling the Pony Patrol Puzzle

From Solving to Proving to Designing Puzzles

While presenting puzzles to students and participants at workshops, outreach events and classrooms is a lot of fun and an enriching educational experience, inviting them to design original puzzles gives rise to a unique learning opportunity.

At a teaching session I conducted in a community-supported study centre of a neighbouring locality, I discussed the Knight's Tour puzzle and some of the above variations. I then asked the students to design their own puzzle.

In the course of our discussion, we arrived at a rather ingenious puzzle which entailed a Knight's Tour on the cross-shaped 48-squared board shown in Figure 33.

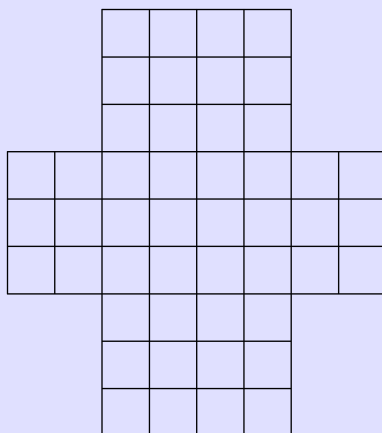


Figure 33: The 48 Squared Puzzle Board

The 48-squared puzzle was an extension of the 3×4 board puzzle. We observed that if we start from corner square of a 3×4 board, the path ends on an adjacent middle-edge square. Thus, we realized that we could begin on a middle left square and have a path that ends on the bottom-left square. This meant that we can attach a second 3×4 board to the first in such a way that the a knight can jump from the bottom corner square of the first board onto the middle square of the edge of the second board (Figure 34). From here the knight can complete a tour of the second board.

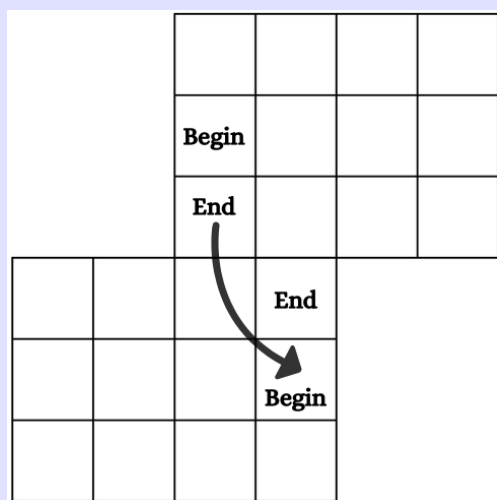


Figure 34: A second 3×4 board attached

After attaching the second board, we realized that a third 3×4 board could be attached adjacent to the second board so that the knight could hop from the end square of its second-board journey onto a corner square of the third 3×4 board, (as shown in Figure 35) and complete its journey.

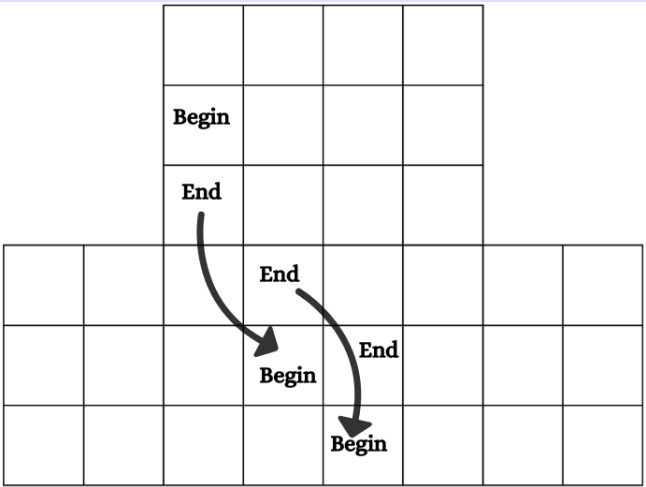


Figure 35: A second 3×4 board attached

Soon we realized that in fact a fourth 3×4 board could be attached from where the knight could jump from the third board such that the four 3×4 boards formed a symmetric shape.

We could now confidently predict a knight’s tour where the 1st, 12th, 13th, 24th, 25th, 36th, 37th and 48th squares visited by the knight would be as shown in Figure 36.

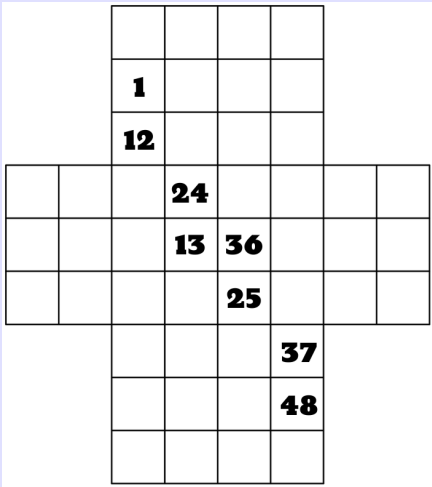


Figure 36: Predicted Solution of Knight’s Tour on 48-square Board

Indeed as we explored the puzzle, we found that our hypothesis was correct, and we had found for ourselves an original puzzle with a verified solution (Figure 37)!

		10	7	2	5		
		1	4	11	8		
		12	9	6	3		
15	18	21	24	27	30	35	32
20	23	16	13	36	33	26	29
17	14	19	22	25	28	31	34
		46	43	40	37		
		41	38	45	48		
		44	47	42	39		

Figure 37: The Solution of the Final Puzzle

Conclusion/Exercise

The above heuristic approach helped me design quite a few Knight's Tour puzzles myself, and I encourage my readers to try out the same. Here is an exercise-puzzle for the reader:

Find a Knight's tour path for a 6×8 rectangular chequerboard.

Hint: A 6×8 chequerboard comprises four (conveniently positioned) 3×4 chequerboards! How many such paths can you find using the above heuristic? Can you find other Knight's tour paths?

Further reading

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About the author

ROSSI D'SOUZA is an Assistant Professor of Physics at the Chhatrapati Shivaji Maharaj Institute of Technology, Panvel, Mumbai. He has a PhD in Mathematics Education from Homi Bhabha Centre for Science Education, TIFR. He has a wide range of experience in teaching at the college and university level and in science communication. His research interests range from developing mathematical puzzles and exploring their pedagogical potential and philosophical implications to uncovering the political economy of mainstream STEM education research at the international level. He may be contacted at rossi.dsouza@gmail.com.

Symmetric Quadrilaterals: Do We Know Them?

Article contributed by Swati Sircar

If we consider the different types of quadrilaterals — square, rectangle, rhombus, parallelogram, kite, trapezium, isosceles trapezium, dart (or non-convex kite) etc. — we can easily state the kind of symmetry of each type. For example, rhombi and rectangles have line symmetry and rotational symmetry of order 2.

But what happens when we flip the question?

The following questions are addressed initially to the student. We invite you (the student) to explore these questions as thoroughly as possible.

In the latter part of the article, we address our remarks to the teacher.

Questions for the student

- (A) Given that a quadrilateral has line symmetry, what kind of quadrilateral must it be?
- (B) Given that a quadrilateral has rotational symmetry, what kind of quadrilateral must it be?

The answer to the second question is well known. (You may be able to guess the solution.)

But the first question is much less explored. We examine the possibilities.

(A) Quadrilaterals with line symmetry

1. Given that a quadrilateral has line symmetry, the line of symmetry can pass through a vertex or it can intersect a side. Can it do both? Why or why not?
 - (a) Without loss of generality, let us consider a quadrilateral $ABCD$ such that its line of symmetry passes through vertex A . Clearly, vertices B and D must lie on either side of the line of symmetry. Why?
 - (b) So, where should vertex C be? Why?
 - (c) Now what can we say about AB and AD ? What about BC and DC ? So, what kind of quadrilateral is $ABCD$?
2. Now if quadrilateral $PQRS$ is such that its line of symmetry intersects PQ at some point other than P or Q , then does it pass through R or S ? Why?

- (a) Clearly P and Q are on either side of the line of symmetry.
Where are R and S relative to this line?
- (b) What can we say about PQ and RS ?
- (c) What can we say about QR and PS ?
- (d) So, what kind of quadrilateral is $PQRS$?

(B) Quadrilaterals with rotational symmetry

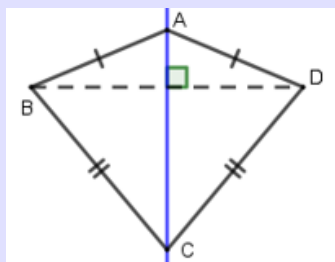
- 3. If a quadrilateral has rotational symmetry, then what is the minimum order (of its rotational symmetry)? Why?
 - (a) Consider a quadrilateral $LMNO$ which has rotational symmetry.
If rotated, to which vertex can L go? What are the possibilities?
(Note that the rotation can be either clockwise or anticlockwise.)
 - (b) Consider the rotation with minimum order and the corresponding angle of rotation. If $LMNO$ is rotated, say anticlockwise, through this angle, what are the new positions of L, M, NO ?
 - (c) Consequently, where do LM and MN go? What can we conclude based on that?
 - (d) So, what kind of quadrilateral is $LMNO$?
- 4. **Optional:** If a polygon has n sides and rotational symmetry, what are its possible orders of rotation?

Notes for the teacher

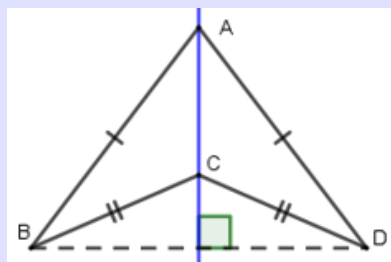
(A) Quadrilaterals with line symmetry

1. Line symmetry through a vertex If quadrilateral $ABCD$ has a line of symmetry through A , then B and D must lie on either side of the line and C must lie on the line itself. (For, if C is not on the line of symmetry, then there must be a 5th vertex E which is symmetric to C in the perpendicular bisector of CE , which acts as the line of symmetry. But $ABCD$ is a quadrilateral and there cannot be a 5th vertex.) So, A and C are on the line of symmetry, while B and D are on opposite sides of this line. Therefore, the line of symmetry must be the perpendicular bisector of BD .

So, $AB = AD$ and similarly, $BC = DC$. Therefore, $ABCD$ is a kite, if convex or a dart, if concave (Figure 38).



(a)



(b)

Figure 38

2. Line symmetry through a side If the line of symmetry of quadrilateral $PQRS$ intersects PQ at a point other than P or Q , then it cannot pass through R or S . For, suppose that the line of symmetry passes through R ;

then by the above, P too would lie on the line of symmetry. But that is a contradiction! For the same reason, the line cannot pass through S . So, R and S must lie on opposite sides of the line of symmetry.

This implies that the line of symmetry is the perpendicular bisector of PQ and SR . Therefore, $PQ \parallel SR$. So, $PQRS$ is a trapezium.

Now P and S are on one side while Q and R are on the other side. (We are not considering crossed polygons.) Therefore, $PS = QR$. It follows that $PQRS$ is an isosceles trapezium (Figure 39).

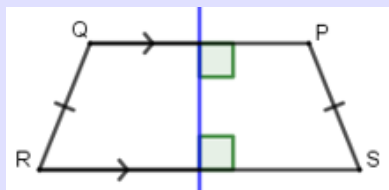


Figure 39

(B) Quadrilateral with rotational symmetry

3. Possible order of rotational symmetry for quadrilateral: If $LMNO$ has rotational symmetry and it is rotated preserving symmetry, then L can rotate to M or O , or to N . If M , then successive rotations would take $L \rightarrow M \rightarrow N \rightarrow O \rightarrow L$, therefore, it would be a rotation of order 4. Similarly, if L rotates to O , then again it would be a rotation of order 4. Two such rotations would bring L to N . On the other hand, if L rotates to N , then successive rotations would result in $L \rightarrow N$, i.e., it would be a rotation of order 2. In either case, L can be rotated to the opposite vertex N . So, the minimum possible order is 2 and angle of rotation $360^\circ / 2 = 180^\circ$.

In such a case, opposite vertices get switched, i.e., $L \leftrightarrow N$ and $M \leftrightarrow O$. So, $LM \leftrightarrow NO$ and $MN \leftrightarrow LO$. Therefore, $LM = NO$ and $MN = LO$, i.e., both pairs of opposite sides are equal. So, $LMNO$ is a parallelogram (Figure 40).

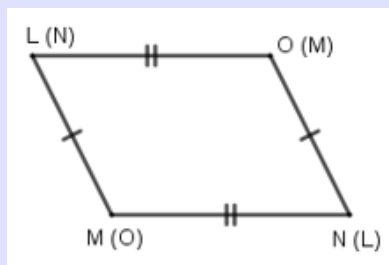


Figure 40

4. Order of a polygon with rotational symmetry and the number of its sides

Let $A_1A_2 \dots A_n$ be a n -sided polygon with rotational symmetry of order n ; see Figure 41. Assume that the vertices are labelled in anticlockwise order.

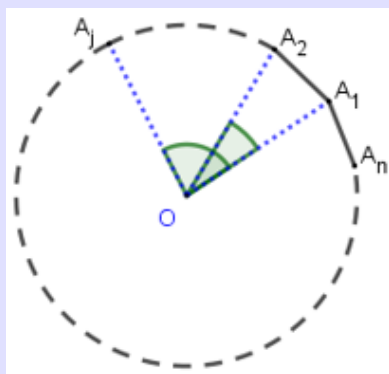


Figure 41

Let O be the centre of rotation. The minimum possible angle of rotation is $360^\circ/n$. For any order k of rotation, the corresponding angle is $360^\circ/k$. Let A_1 go to A_j under a rotation (say anticlockwise) through this angle.

Now, if $A_1 A_2 \dots A_n$ has rotational symmetry of order n , and we rotate it by $360^\circ/n$ anticlockwise, then

$$A_1 \rightarrow A_2, \quad A_2 \rightarrow A_3, \quad \dots \quad A_m \rightarrow A_{m+1}, \quad \dots \quad A_n \rightarrow A_1.$$

So, $j - 1$ successive turns through $360^\circ/n$ each take A_1 to A_j . It follows that

$$\frac{360^\circ}{n} \times (j - 1) = \frac{360^\circ}{k},$$

$$\therefore (j - 1)k = n,$$

hence k is a divisor of n . So, the order of rotation of a polygon is a divisor of the number of its sides.

Remark

If a polygon has rotational symmetry, then the order is more than 1. If a polygon has no rotational symmetry, then the order is 1 which is a factor of every natural number.

Also, if an n -sided polygon has rotational symmetry of order n , then it is a regular polygon. Do you see why?

Therefore, if a polygon with prime number of sides has rotational symmetry, then it is a regular polygon.

About the author

SWATI SIRCAR is the web-editor of *At Right Angles*, published by Azim Premji University. Mathematics is the second love of her life (the first being drawing). She has B.Stat-M.Stat degrees from the Indian Statistical Institute and a MS in mathematics from the University of Washington, Seattle. She has been ‘doing’ mathematics with children and teachers for more than one and a half decades and is deeply interested in anything hands-on — origami in particular. She initiated Math Space, a mathematics laboratory at the Bengaluru campus of Azim Premji University. It caters to schools, teachers, parents, children, NGOs working in school education and teacher educators. Swati may be contacted at swati.sircar@apu.edu.in.

The First Transcendental Number

Article contributed by Sasikumar

The family of complex numbers ($\mathbb{C}$) is vast (and, well, complex). Within it we find so many interesting subfamilies: the real numbers $\mathbb{R}$, the rational numbers $\mathbb{Q}$, the set of integers $\mathbb{Z}$, and so on.

When we study polynomial equations, we discover a natural subdivision of the complex numbers: Algebraic numbers and Transcendental numbers.

Definition of an algebraic number

A number in $\mathbb{C}$ is said to be an *Algebraic Number* if it is a solution of a single-variable polynomial equation with integer coefficients.

Definition of a transcendental number

A number in $\mathbb{C}$ that is not algebraic is said to be *Transcendental*.

At the outset, it may not be obvious that there exist transcendental numbers at all. We will have to show that they do exist. Let us start by exhibiting some examples of algebraic numbers.

Examples of algebraic numbers

1. A rational number is an algebraic number. For, the rational number p/q (where $p, q \in \mathbb{Z}$, $q \neq 0$) is the solution of the linear equation $qx - p = 0$, which has integer coefficients.
2. $\sqrt{2}$ is an algebraic number as it is a solution of the second-degree equation $x^2 - 2 = 0$, which has integer coefficients.
3. Similarly, $3^{1/5}$ and $10^{1/4}$ are algebraic numbers. They are solutions of the polynomial equations $x^5 - 3 = 0$ and $x^4 - 10 = 0$ respectively.
4. $2^{1/2} + 3^{1/3}$ is an algebraic number. To see why, let $a = 2^{1/2} + 3^{1/3}$. Then $a - 2^{1/2} = 3^{1/3}$, hence $(a - 2^{1/2})^3 = 3$, giving

$$a^3 - 3a^2\sqrt{2} + 6a - 2\sqrt{2} = 3,$$

$$\therefore a^3 + 6a - 3 = 3a^2\sqrt{2} + 2\sqrt{2},$$

$$\therefore (a^3 + 6a - 3)^2 = 2(3a^2 + 2)^2.$$

This simplifies to $a^6 - 6a^4 - 6a^3 + 12a^2 - 36a + 1 = 0$. It follows that a is a root of the polynomial equation

$$x^6 - 6x^4 - 6x^3 + 12x^2 - 36x + 1 = 0.$$

Observe that the polynomial has integer coefficients.

5. $2^{1/2} + 5^{1/3}$ is an algebraic number. We invite you to verify that if $x = 2^{1/2} + 5^{1/3}$, then

$$x^6 - 6x^4 - 10x^3 + 12x^2 - 16x + 17 = 0.$$

6. $\sqrt{2} + \sqrt{3} + \sqrt{5}$ is an algebraic number. We invite you to verify that if $x = \sqrt{2} + \sqrt{3} + \sqrt{5}$, then

$$x^8 - 40x^6 + 352x^4 - 960x^2 + 576 = 0.$$

The algebraic numbers are countable

As can be anticipated from these examples, one can construct an infinite number of irrational numbers which are algebraic (e.g., by using the prime numbers). But this infinity is of a special kind: *the algebraic numbers form a countable set*. This means that it is possible to exhibit a 1–1 correspondence between the set of algebraic numbers and the set of positive integers. This may sound extremely surprising when one first hears it, but it is true, and it can be proved without too much difficulty.

In particular, this means that the set of real numbers that are algebraic is countable.

Now you may be familiar with the fact that the set of real numbers is not countable. This means that any attempt to exhibit a 1–1 correspondence between the set of real numbers and the set of positive integers is doomed to failure. The proof that this is so was first given by George Cantor.

This obviously implies that *there are uncountably many real numbers that are not algebraic*.

In other words, *there are uncountably many real numbers that are not the roots of any single-variable polynomial with integer coefficients.* These are the *transcendental numbers*.

This makes it sound as though it ought to be easy to exhibit examples of real numbers that are transcendental. Here we meet our second surprise. Proving that a given number is transcendental can be very challenging. This happens to be the case for many familiar constants of mathematics that arise naturally in various contexts; e.g., π and e .

Joseph Liouville first proved the existence of transcendental numbers in 1844 (this was before Cantor proved his results about the countable nature of the rational numbers and the algebraic numbers) and gave the first decimal example, the number

$$\sum_{n=1}^{\infty} 10^{-n!} = \frac{1}{10^1} + \frac{1}{10^2} + \frac{1}{10^6} + \frac{1}{10^{24}} + \cdots$$

This number is now known as the *Liouville constant*. It thus has the honour of being the “first transcendental number.”

In 1873 Charles Hermite showed that e is transcendental, and in 1882 Ferdinand Lindemann proved that π too is transcendental; see [1]. The transcendence of π proves conclusively that the ancient problem, first posed by the Greeks, of “squaring the circle” is not solvable. This is the problem of constructing a square equal in area to a given circle, using only Euclidean instruments (a compass and an unmarked straight edge).

Much later, in 1934, A O Gelfond proved that a^b is transcendental whenever a is an algebraic number other than 0 and 1 and b is an irrational algebraic number. This is a very powerful result that allows us to generate as many transcendental numbers as we may want. For example, the number $2^{\sqrt{2}}$.

As a corollary, it implies that e is transcendental. For, we have Euler's result, $e^{i\pi} = -1$. If e were algebraic, then Euler's result would contradict Gelfond's theorem, since $i\pi$ is irrational. Hence e must be transcendental.

We will not prove any of these results here. (*Note from the editor:* The proofs are beyond the scope of this magazine. We remark in passing that in the case of π , it is challenging even to prove that it is an irrational number. On the other hand, it is easy to prove that e is irrational. The proof that π is irrational was first given by Lambert in 1768.)

Further reading

[1] Wikipedia, "Transcendental number" from

https://en.wikipedia.org/wiki/Transcendental_number

About the author

SASIKUMAR K is presently working as a PG Teacher in Mathematics at Jawahar Navodaya Vidyalaya, North Goa. He completed his M Phil under the guidance of Dr.K.S.S. Nambooripad. Earlier, he worked as a maths Olympiad trainer for students of Navodaya Vidyalaya Samiti in Hyderabad. He has research interests in Real Analysis and Commutative Algebra. He may be contacted at 112358.ganitham@gmail.com

A Corollary of the Weak Goldbach Conjecture

Article contributed by Sourajit Mandal

Introduction

The Weak Goldbach Conjecture (WGC), proved by Harald Helfgott in 2013, states that every odd integer greater than five can be expressed as the sum of three prime numbers.

We prove a corollary of the WGC that reveals a relationship between the factorization of a natural number and its additive partitions: whenever N has $h > 0$ prime factors exceeding 5, it admits at least $2^h - 1$ partitions in which every summand has the same total number of prime factors (counted with multiplicity) as N itself.

Background

The *Goldbach Conjecture* (1742) asserts that every even integer greater than 2 is the sum of two primes; it remains open.

Its weaker variant, known as the *Weak Goldbach Conjecture* (WGC), [5], asserts that every odd integer greater than 5 is the sum of three primes.

After more than 250 years, Helfgott [3, 4] provided a complete proof in 2013, combining deep theory with extensive computational verification. The WGC is therefore now regarded as a theorem.

Main Result

Corollary (Corollary of the WGC). *Let N be a positive integer with prime factorization*

$$N = 2^{k_1} 3^{k_2} 5^{k_3} \cdot p_1 p_2 \cdots p_h$$

where $p_1, p_2, \dots, p_h$ are distinct primes each greater than 5, and $k_i \geq 0$. Let k denote the sum $k_1 + k_2 + k_3 + h$. Then:

1. *N has at least $2^h - 1$ additive partitions such that every summand in each partition has exactly k prime factors (counted with multiplicity).*
2. *For each integer r with $1 \leq r \leq h$, there exist at least $\binom{h}{r}$ such partitions whose summands number exactly 3^r .*

Examples

Example 1 ($h = 1$). Let $N = 42 = 2 \cdot 3 \cdot 7$, so $k = 3$ and $h = 1$ (only $7 > 5$). By the WGC, $7 = 2 + 2 + 3$. Substituting:

$$42 = 2 \cdot 3 \cdot (2 + 2 + 3) = (2 \cdot 3 \cdot 2) + (2 \cdot 3 \cdot 2) + (2 \cdot 3 \cdot 3) = 12 + 12 + 18.$$

Each summand has $k = 3$ prime factors. We expect at least one partition with $3^1 = 3$ terms (since $2^1 - 1 = 1$). This is the case.

Example 2 ($h = 2$). Let $N = 770 = 2 \cdot 5 \cdot 7 \cdot 11$, so $k = 4$ and $h = 2$ (both $7, 11 > 5$). We expect $2^2 - 1 = 3$ partitions: two with $3^1 = 3$ terms and one with $3^2 = 9$ terms. This is what we do find.

Partition 1 (expand $7 = 2 + 2 + 3$):

$$770 = 2 \cdot 5 \cdot (2 + 2 + 3) \cdot 11 = 220 + 220 + 330.$$

Partition 2 (expand $11 = 3 + 3 + 5$):

$$770 = 2 \cdot 5 \cdot 7 \cdot (3 + 3 + 5) = 210 + 210 + 350.$$

Partition 3 (expand both 7 and 11):

$$\begin{aligned} 770 &= 2 \cdot 5 \cdot (2 + 2 + 3) \cdot (3 + 3 + 5) \\ &= 60 + 60 + 100 + 60 + 60 + 100 + 90 + 90 + 150. \end{aligned}$$

Every summand carries $k = 4$ prime factors.

Proof of Corollary

We have, $N = 2^{k_1} 3^{k_2} 5^{k_3} \cdot p_1 p_2 \cdots p_h$, where $p_1, p_2, \dots, p_h$ are distinct primes each greater than 5, and $k_i \geq 0$.

Step 1: Applying the WGC to each large prime. Since every prime $p_i > 5$ is an odd integer greater than 5, by the WGC there exist primes a_i, b_i, c_i such that

$$p_i = a_i + b_i + c_i.$$

Step 2: Forming partitions by choosing primes to expand. Consider all non-empty subsets $S \subseteq \{p_1, \dots, p_h\}$. There are $2^h - 1$ such subsets. For a subset S of size r , replace each prime in it by its three-prime sum; then distribute the product over the resulting sums. On expanding the product we get 3^r terms. Each term is a product of k primes, as claimed.

Step 3: Counting the partitions. The number of non-empty subsets of $\{p_1, \dots, p_h\}$ of size r is $\binom{h}{r}$. Each such subset yields a partition with 3^r summands. Summing over all subset sizes,

$$\sum_{r=1}^h \binom{h}{r} = 2^h - 1,$$

giving $2^h - 1$ distinct partitions in all.

Since the construction yields at least $\binom{h}{r}$ partitions with 3^r terms for each r , and at least $2^h - 1$ partitions overall, the corollary is proved. ■

Remarks

- It is interesting to note how the WGC can be used to obtain a result about integer partitions.
- The lower bound proved above is weak; the actual number of such partitions is bound to be much larger. However, obtaining tighter bounds is a difficult problem.

About the author

SOURAJIT MANDAL is a Grade 12 student at MatriKiran High School, Gurgaon, India. His academic interests are primarily in Mathematics, Physics and Computer Programming. From an early age, he has had a fascination with Astronomy. He began writing articles for Space India's magazine *Galactica* in Grade 7 and is now the author of *The Wonders of Cosmos*, published by Space India.

Sourajit may be contacted at sourajitmandal09@gmail.com

Further reading

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Part VI

Culture & History of Mathematics

Katherine Johnson: The Mathematician behind the Moon Landing

Katherine Coleman Goble Johnson was a pioneering NASA African American mathematician whose precise manual calculations of orbital mechanics were critical to the success of the first U.S. crewed lunar mission. She was an outlier whose life reflected an indomitable spirit, one that overcame obstacles through sheer brilliance and devotion to computation.

Johnson's legacy extends much beyond a historical milestone. What is striking is the manner in which she was able to rise above the constraints of gender and racial discrimination — not via protest but through the elegance and precision of her mathematics.

The Trajectory of a Mathematical Prodigy

Born in 1918 in White Sulphur Springs, a quiet town of West Virginia, USA, Katherine Coleman experienced life mainly through the lens of numbers. From an early age, she exhibited a curiosity and articulateness that set her apart. Inspired by a father who possessed an intuitive aptitude for mental arithmetic despite a limited formal education, and a mother who was a nurturing educator, Katherine's innate love for mathematics blossomed naturally. She counted everything — the steps to her house and church; the forks and plates she washed. This simple act was the start of a trajectory that would take her to NASA and help map humanity's path to the stars.

Educational opportunities for Black children in her hometown ended at the eighth grade. Her family's commitment to education forced them to find a way around this obstacle. Every autumn, her mother moved with the children 125 miles to Institute, West Virginia, so they could attend high school and college, while her father remained behind to work and fund their schooling.

Katherine was a prodigy; she was ready for high school by the age of ten and graduated at fourteen. Her college years at West Virginia State College (WVSC) were transformative. She took every single mathematics course the college offered. Her talent caught the attention of brilliant mentors. Angie Turner King, who had taught her high school geometry, continued to mentor her in college. James Carmichael Evans, an MIT graduate, steered her toward mathematics major despite her equal prowess in French.

However, her most academic relationship was with Dr. W.W. Schieffelin Claytor, the third African American to ever earn a Ph.D. in mathematics. Recognizing her potential, Claytor told her she would make a great

research mathematician and took it upon himself to prepare her for that future. He designed advanced courses for her, including one in analytic geometry. This would later become the foundation of her contributions to space exploration. She graduated with honours at the age of eighteen with degrees in mathematics and French.

The Era of ‘Human Computers’

When Johnson graduated from college, the US was still “segregated”. Different races were separated from each other in many places and activities. African-Americans were rarely able to have jobs in mathematics and science. It was also unusual for women (of any race) to have degrees in mathematics. The only professional job available to Johnson after graduation was teaching. She taught in public schools for many years but stopped when she married and had children.

Her true calling took form in 1952 when she learned of opportunities for Black female mathematicians at the National Advisory Committee for Aeronautics (NACA) Langley Laboratory. In 1953, at the age of 34, she joined the West Area Computing Section — an all-Black group of skilled female mathematicians known as “human computers” — where she initially verified wind tunnel data and aircraft trajectories.

In a rigidly segregated Virginia, Black women were confined to separate workspaces, dining areas, and even restrooms. But Katherine Johnson’s abilities could not be kept confined to such boundaries. Soon after she began her tenure, she was temporarily assigned to a previously all-White, all-male section of the Flight Research Division.



Figure 42: Katherine Johnson, mid-1960s at NASA. Credit: NASA Langley Research Centre, Hampton, Virginia.

Women were excluded from meetings with male engineers and scientists, by policy. Determined both to learn and to contribute, Johnson insisted on attending meetings. Known for her strong grounding in geometry, she began working alongside teams of men. There, she asked sharp, probing questions, pushed her way into high-level briefings where the challenges of spaceflight were being examined, and displayed a command of analytic geometry that left even seasoned engineers deeply impressed.

In 1958, NACA officially became NASA. She never returned to the computing pool. She operated under a simple but profound personal philosophy instilled by her father — *she was as good as anybody, but never better* — and could navigate the prejudices of the era with grace, dignity and intellectual authority.

Transition into ‘Machine Computing’

Katherine Johnson’s career at NASA paralleled one of the most significant technological transformations of the 20th century — the shift from human, hand-driven computation to electronic data processing. As the Space Race intensified, NASA began relying on massive, room-sized IBM computers to perform the complex calculations required for orbital mechanics.

These early computers lacked something crucial: human trust! Astronauts, whose lives depended entirely on the accuracy of these calculations, were wary of the untested “black boxes” producing numbers they could not independently verify. This skepticism reached a turning point before astronaut John Glenn’s historic 1962 orbital mission on Friendship 7. Although the IBM computer had generated the capsule’s trajectory, Glenn refused to proceed till “that girl” — referring specifically to Katherine Johnson — checked the numbers and the orbit by hand.

Johnson used her mechanical calculating machine along with tables of logarithms and trigonometric functions to reconstruct the trajectory. She calculated each stage with care, balancing factors such as Earth’s rotation, the capsule’s velocity and subtle gravitational variations. Her results matched the computer’s output exactly. Johnson’s validations did more than just clear John Glenn for lift-off; they established NASA’s confidence in the new era of digital computing technology.

The Beauty and Complexity of Katherine Johnson's work at NASA

The true measure of Katherine Johnson's legacy lies in the complexity of the problems she solved at NASA. Orbital mechanics is a complex interplay of non-linear equations, requiring working with multiple variables. What appears as the smooth motion of a spacecraft is the outcome of intricate mathematics.

This legacy may be seen in the 1960 NASA report she co-authored with T. H. Skopinski, *Determination of Azimuth Angle at Burnout for Placing a Satellite over a Selected Earth Position*. At a time when female "computers" were rarely credited as authors, Johnson's name appearing on the paper marked a major shift. Having carried out much of the work, both theoretical and computational, she had been acknowledged — breaking an important gender barrier within NASA.

The report is a sophisticated work of applied mathematics. It establishes the framework for linking a satellite's position to its projected latitude and longitude on a rotating Earth. The central challenge was to determine the exact azimuth angle — the compass direction from true north — at which a rocket's engines must cut off to ensure that, after a specified number of orbits, the spacecraft would pass precisely over a designated point on Earth.

To solve this, Johnson used an iterative method. Starting with an initial approximation, she refined the calculations till the results converged. Her work also accounted for the Earth's true shape as an oblate spheroid (slightly flattened at the poles), a correction that affects gravitational behaviour and orbital paths.

This work proved to be vital for early American space missions, e.g., for Alan Shepard's 1961 suborbital flight, ensuring accurate recovery operations at sea. Johnson's ability to mentally navigate such problems was legendary: she would work backward from a desired landing point to determine the precise moment of launch.

Her expertise extended beyond Earth's orbit. During the Apollo 11 Moon Landing, in 1969, she contributed to calculations that ensured the successful rendezvous between the Lunar Module and the Command Module orbiting the Moon — a task requiring great precision, where even a tiny miscalculation could send the module drifting endlessly into empty space.

Perhaps her most critical, life-saving application of orbital mathematics occurred during the aborted Apollo 13 mission in 1970. When an oxygen tank explosion crippled the spacecraft and knocked out its primary digital navigation and life support systems, the crew was forced to rely on backup procedures. Years earlier, Johnson had anticipated the possibility of a total electrical failure and had helped develop backup navigational charts based on a specialized one-star observation system. By using her pre-calculated charts and star sightings with a mechanical sextant, the astronauts were able to determine their precise attitude and position, executing a crucial manual engine burn that returned them safely back to Earth. Through such work, Johnson transformed abstract mathematics into life-saving precision.

Legacy

Katherine Johnson authored more than 26 scientific reports over her 33-year career at NASA. Her work spanned major initiatives, e.g., the Space Shuttle program, the Earth Resources Satellite, and studies for Mars missions.

She retired in 1986, leaving behind a legacy of excellence. In 2015, she was awarded the Presidential Medal of Freedom in recognition of her many contributions to spaceflight. In 2016, NASA Langley Research Center dedicated a facility in her name: the Katherine G. Johnson Computational Research Facility. The same year, she also received honorary doctorates from West Virginia University and West Virginia State University.

In her retirement years, she enjoyed travelling, playing bridge and spending time with family and friends. She remained deeply connected to education, often engaging with students and encouraging their curiosity.

She passed away in 2020 at the age of 101. Her life was never driven by the pursuit of fame or the mere act of breaking barriers, but by a steadfast commitment to solving problems with clarity and unwavering dedication.

Reflecting on her extraordinary journey, her own words capture the essence of her spirit: "My problem was to answer questions. And I did that to the best of my abilities at all times. And it was a joy!"

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Acknowledgment: The following sources were used to generate summaries with *Gemini AI*. These were subsequently compiled into this essay.

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3. "Katherine Johnson biography." NASA. Available at: <https://www.nasa.gov/content/katherine-johnson-biography>.
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Further reading

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- [2] *Reaching for the Moon – The autobiography of NASA mathematician Katherine Johnson* by Katherine Johnson.
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Article contributed by a friend of the editors

Part VII

Miscellaneous Articles – 2

Grandma's Identities – II

In the January 2026 issue of *The Compass*, we described a trigonometric identity playfully called *Grandma's Identity*. Not only was the identity very beautiful, but the method used to prove it was equally striking and noteworthy. We describe another identity of this kind in this article.

Recap: In the January 2026 article, we used the following algebraic identity which is true for all odd positive integers n ,

$$x^n - 1 = (x - 1) \prod_{k=1}^{(n-1)/2} \left(x^2 - 2x \cos \frac{2\pi k}{n} + 1 \right),$$

to prove that

$$\prod_{k=1}^{(n-1)/2} 2 \cos \frac{\pi k}{n} = 1.$$

This is the 'Grandma identity.'

A twin sister

In [2], Sury mentions that ‘Grandma’s identity’ has a ‘twin sister,’ namely:

$$\prod_{s=1}^{2k} 2 \cos \frac{2\pi s}{4k+1} = (-1)^k = \prod_{s=1}^{2k-1} 2 \cos \frac{2\pi s}{4k-1}.$$

He writes, “We call this the twin sister because it is seen to be equivalent to the Grandma identity on using the formula $\cos(\pi - \theta) = -\cos \theta$.”

Here are some instances of the twin sister identity:

$$2 \cos \frac{2\pi}{3} = -1,$$

$$2 \cos \frac{2\pi}{5} \cdot 2 \cos \frac{4\pi}{5} = -1,$$

$$2 \cos \frac{2\pi}{7} \cdot 2 \cos \frac{4\pi}{7} \cdot 2 \cos \frac{6\pi}{7} = 1,$$

$$2 \cos \frac{2\pi}{9} \cdot 2 \cos \frac{4\pi}{9} \cdot 2 \cos \frac{6\pi}{9} \cdot 2 \cos \frac{8\pi}{9} = 1,$$

etc.

We leave the task of explaining Sury’s remark to the reader. In what way does the identity $\cos(\pi - \theta) = -\cos \theta$ explain the twin sister?

A distinguished Italian cousin

Sury then points out that there is a closely related identity, which he calls an “Italian cousin.”

This is the following identity which is true for all odd positive integers n :

$$\prod_{r=1}^{(n-1)/2} \left(3 + 2 \cos \frac{2\pi r}{n} \right) = F_n.$$

Here F_n is the n -th number in the Fibonacci sequence

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots$$

This relation certainly puts the Grandma identity into very distinguished company!

Older sister of the Italian cousin

There is a slightly more general result than the statement quoted above:

Theorem. *Let $n > 1$ be an integer. Let*

$$f(n) = \prod_{r=1}^{\lfloor n/2 \rfloor} \left(3 + 2 \cos \frac{2\pi r}{n} \right).$$

Then $f(n) = F_n$, the n -th Fibonacci number.

Let us verify the theorem for a few small values of n . For $n = 1, 2$, there is nothing to verify.

n = 3: We have,

$$f(3) = 3 + 2 \cos \frac{2\pi}{3} = 3 - 1 = 2,$$

so $f(3) = F_3$ (verified).

n = 4: We have,

$$f(4) = \left(3 + 2 \cos \frac{2\pi}{4} \right) \cdot \left(3 + 2 \cos \frac{4\pi}{4} \right) = 3 \cdot 1 = 3,$$

so $f(4) = F_4$ (verified).

n = 5: We have,

$$\begin{aligned} f(5) &= \left(3 + 2\cos\frac{2\pi}{5}\right) \cdot \left(3 + 2\cos\frac{4\pi}{5}\right) \\ &= \frac{5 + \sqrt{5}}{2} \cdot \frac{5 - \sqrt{5}}{2} \\ &= \frac{20}{4} = 5, \end{aligned}$$

so $f(5) = F_5$ (verified).

n = 6: We have,

$$\begin{aligned} f(6) &= \left(3 + 2\cos\frac{2\pi}{8}\right) \cdot \left(3 + 2\cos\frac{4\pi}{6}\right) \cdot \left(3 + 2\cos\frac{6\pi}{6}\right) \\ &= 4 \cdot 2 \cdot 1 = 8, \end{aligned}$$

so $f(6) = F_6$ (verified).

The proof of this remarkable identity is more involved than one may expect. We will postpone it to the next article of this series. But we list the main steps of the proof here. The reader could try filling in the missing details.

Step 1. Invoke the “Binet formula” for the n -th Fibonacci number:

$$F_n = \frac{u^n - v^n}{u - v},$$

where u, v are the roots of the quadratic equation $z^2 = z + 1$ or $z^2 - z - 1 = 0$:

$$u = \frac{1 + \sqrt{5}}{2}, \quad v = \frac{1 - \sqrt{5}}{2}.$$

Step 2. Recall the relations $u + v = 1$ and $uv = -1$ which imply that $u^2 + v^2 = 3$ and $u^2v^2 = 1$. So u^2, v^2 are the roots of the quadratic equation $z^2 - 3z + 1 = 0$, and we have,

$$z^2 - 3z + 1 = (z - u^2)(z - v^2), \quad z^2 + 3z + 1 = (z + u^2)(z + v^2).$$

Step 3. Recall that the numbers L_n defined by

$$L_n = u^n + v^n$$

yield the *Lucas sequence*: 1, 3, 4, 7, 11, 18, 29, 47, The Lucas sequence is intricately connected with the Fibonacci sequence.

Step 4. Write down the complete factorization of $3 + 2\cos\theta$ in terms of the n -th roots of unity.

Step 5. Pair up the factors suitably and simplify to the extent possible.

After resolutely following each step to its logical conclusion, we arrive at the stated identity.

in [2], Sury follows a different approach, starting with an algebraic identity. We will describe both these approaches in the next issue. ■

Further reading

[1] Steve Humble, "Grandma's Identity" from *The Mathematical Gazette*, Nov., 2004, Vol. 88, No. 513 (Nov., 2004), pp. 524-525,
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A Parabola & Circle Theorem

We describe here an unexpected property shared by quadratic functions.

The setting: Let a, b be real numbers satisfying the two conditions noted below. Consider the parabola $\mathcal{F}$ with equation $y = x^2 + ax + b$. Assume that $\mathcal{F}$ does not pass through the origin and intersects the x -axis at two distinct point P, Q . These conditions will hold provided $b \neq 0$ and $a^2 > 4b$. Let R be the point where $\mathcal{F}$ intersects the y -axis. The stated conditions then imply that P, Q, R are three distinct non-collinear points.

Let k be the circle passing through P, Q, R . (The circle is uniquely defined.)

Claim: Circle k passes through the point $S(0, 1)$. This is true regardless of the specific values of a, b . We only require that $b \neq 0, a^2 > 4b$.

Please refer to Figure 43 for a sketch. The circle is shown dashed.

Proof of the claim: As we shall see below, the proposition has a very short proof. Indeed, the proof is shorter than the problem statement!

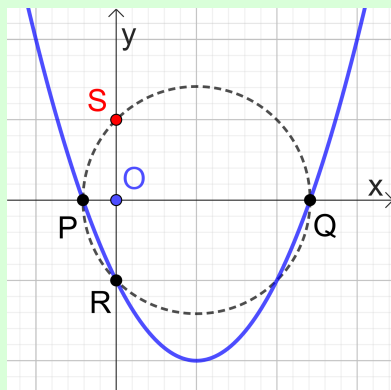


Figure 43: Sketch of the configuration when $a = -2, b = -1$

Referring again to the figure, imagine that the circle has not been drawn. Let S be the point $(0, 1)$. Let us prove that the points P, Q, R, S are concyclic. Note that this is equivalent to the stated proposition. We now have,

$$P, Q, R, S \text{ concyclic} \iff OR \cdot OS = OP \cdot OQ.$$

Since $S = (0, 1)$ and $R = (0, b)$,

$$OR \cdot OS = 1 \cdot b = b.$$

Next, OP, OQ are the roots of the quadratic equation $x^2 + ax + b = 0$, hence

$$OP \cdot OQ = \frac{b}{1} = b \text{ (by Vieté's rule).}$$

Equality thus holds, and it follows that the four points are concyclic. ■

Article contributed by the editors

Call for Articles

We invite mathematics teachers and practitioners to contribute short articles to The Compass.

Articles could deal with mathematical content or the teaching of mathematics. They could feature real-life classroom experiences with projects and explorations.

They could be about the use of audio-visual material or social media in the mathematics classroom.

They could feature reflections on the teaching of mathematics, or on the nature of mathematics.

Articles about the use of software are particularly welcome, as are articles dealing with non-routine problems and classroom experiences with problem-solving.

In short, anything at all to do with school mathematics!